

Department of Computer Science And Engineering (CSE) BRAC University

CSE 251 – Electronic Devices and Circuits

Lecture: 02

Operational Amplifier (OP-AMP)

Introduction and Open Loop Configuration

PREPARED BY

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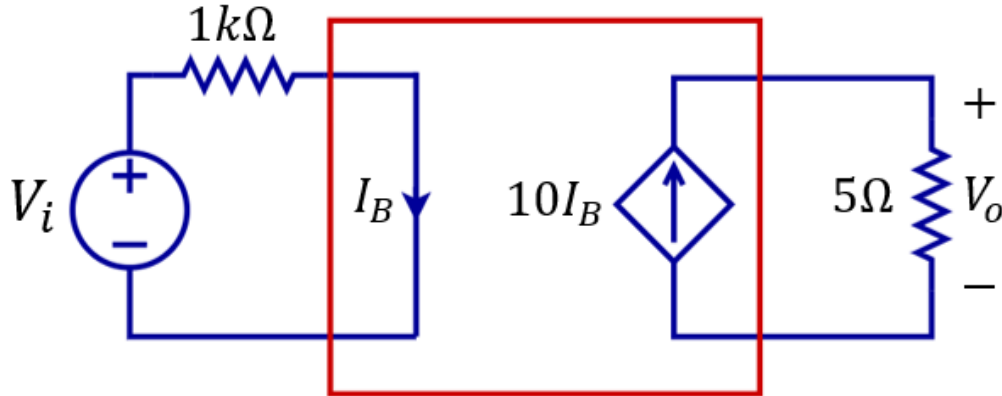
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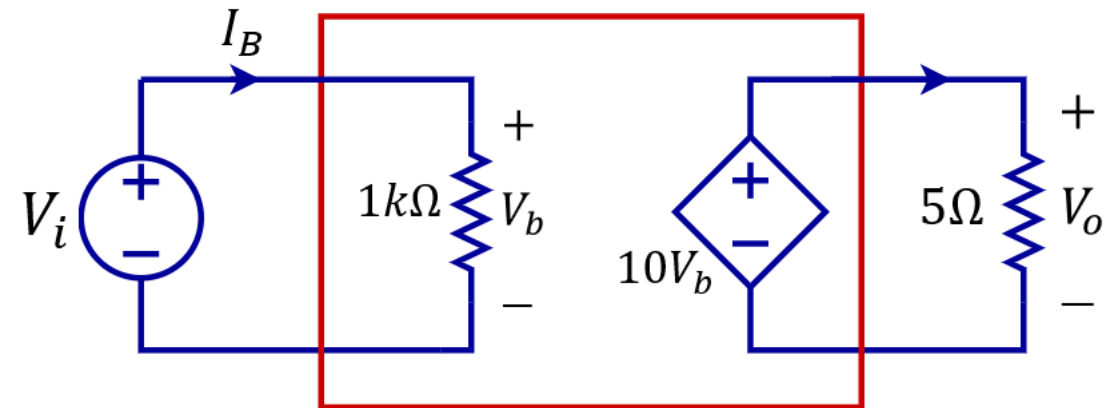
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Dependent Source (Review)

- ❑ To analyze Operational Amplifiers (Op-Amps), we must understand **Dependent Source**.
- ❑ A dependent source generates a **voltage/current** whose value depends on **another voltage or current from a different branch**.
- ❑ Technically, they are **non-linear** – because in real life, dependent sources are made using non-linear electronic devices (e.g. MOSFET, which will be discussed later in this course).
- ❑ However, analysis are similar to linear circuits (CSE250).

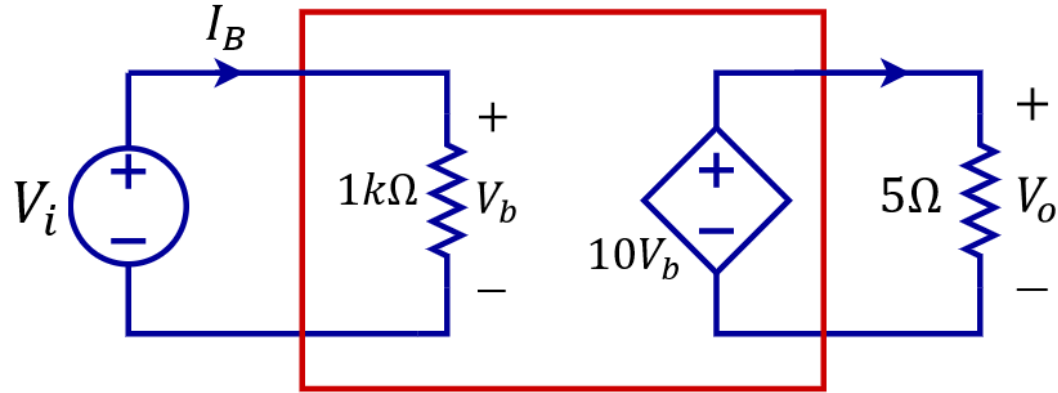


Example: Current Controlled Current Source



Example: Voltage Controlled Voltage Source

Dependent Source (Review)



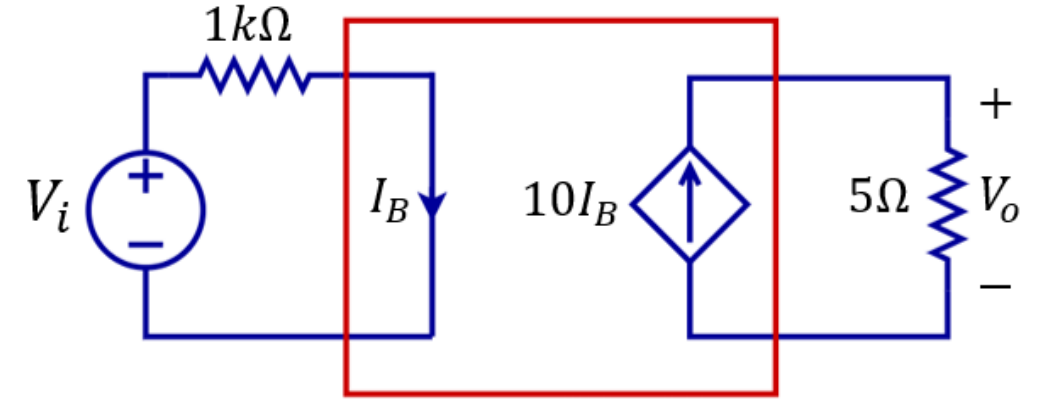
The Voltage of the dependent source depends on the voltage drop across the $1k\Omega$.

$$V_b = I_B \times 1k\Omega = V_i$$

$$V_o = 10V_b = 10V_i$$

Volatge Gain,

$$\frac{V_o}{V_i} = 10$$



The Current of the dependent source depends on the current flow through the $1k\Omega$.

$$I_B = \frac{V_i}{1k\Omega}$$

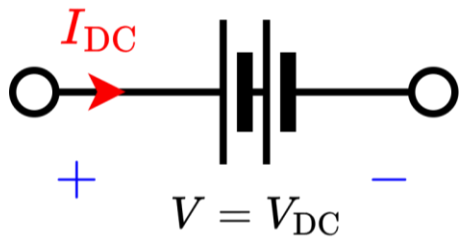
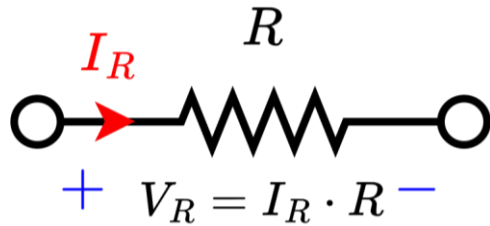
$$\begin{aligned} V_o &= 5\Omega \times 10I_B \\ &= \frac{5 \times 10 \times V_i}{1000\Omega} \\ &= \frac{1}{20} V_i \end{aligned}$$

Volatge Gain,

$$\frac{V_o}{V_i} = \frac{1}{20}$$

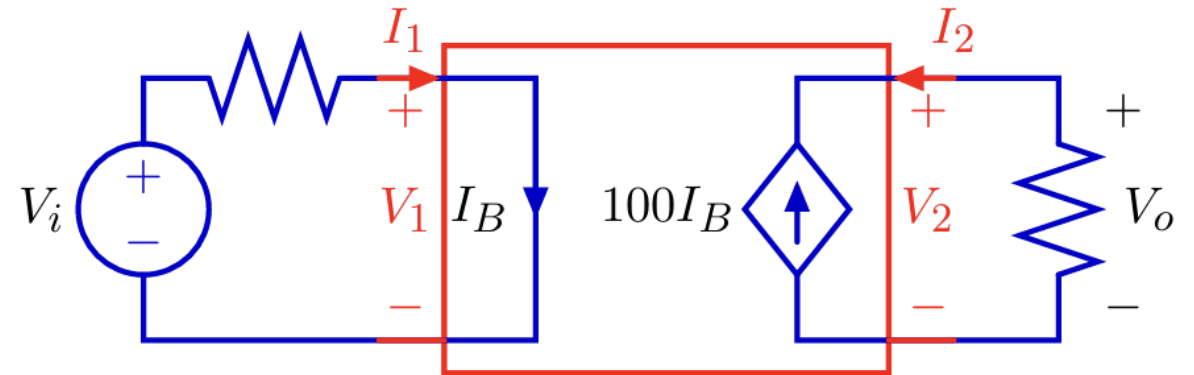
Major Difference

Resistors and Independent sources are single “**port**” device. They are characterized by a **single equation**.



2 terminals, 1 port

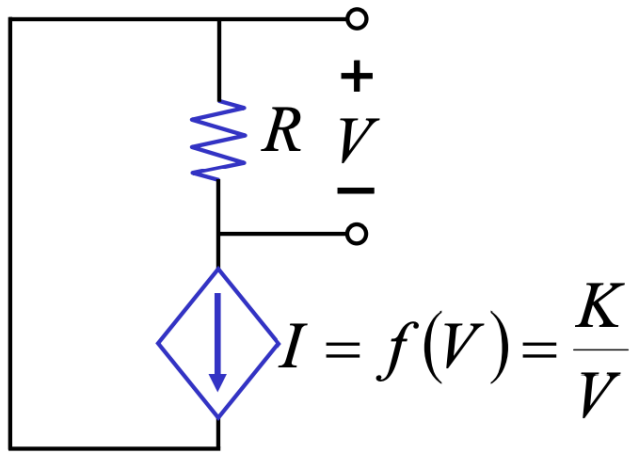
Dependent sources are “**two-ports**” device: characterized by **two equations**.



Here, $V_1 = 0$ and $I_2 = -100I_1$

2 terminals, 2 port

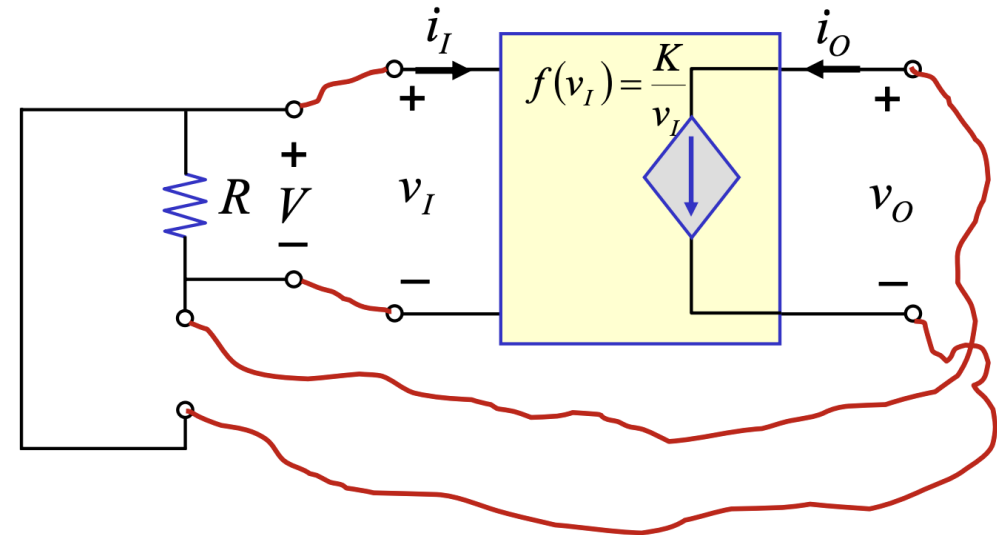
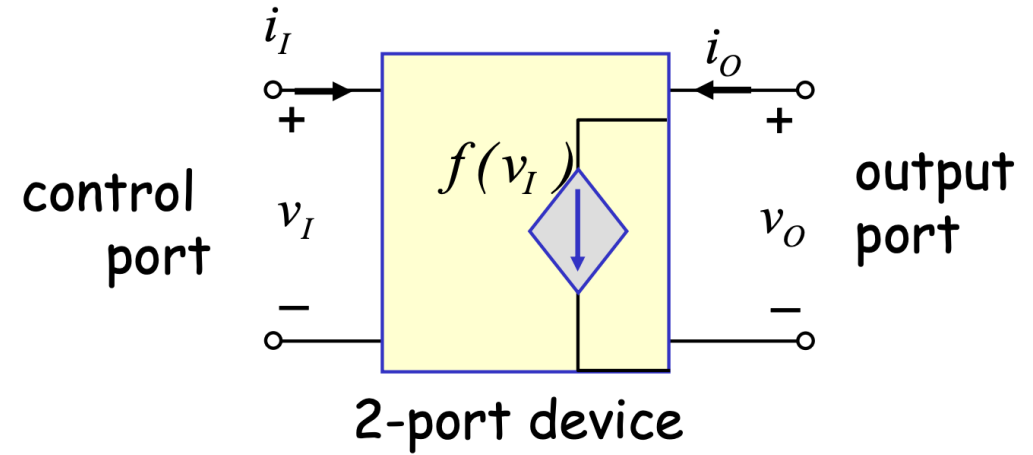
Example



Find V

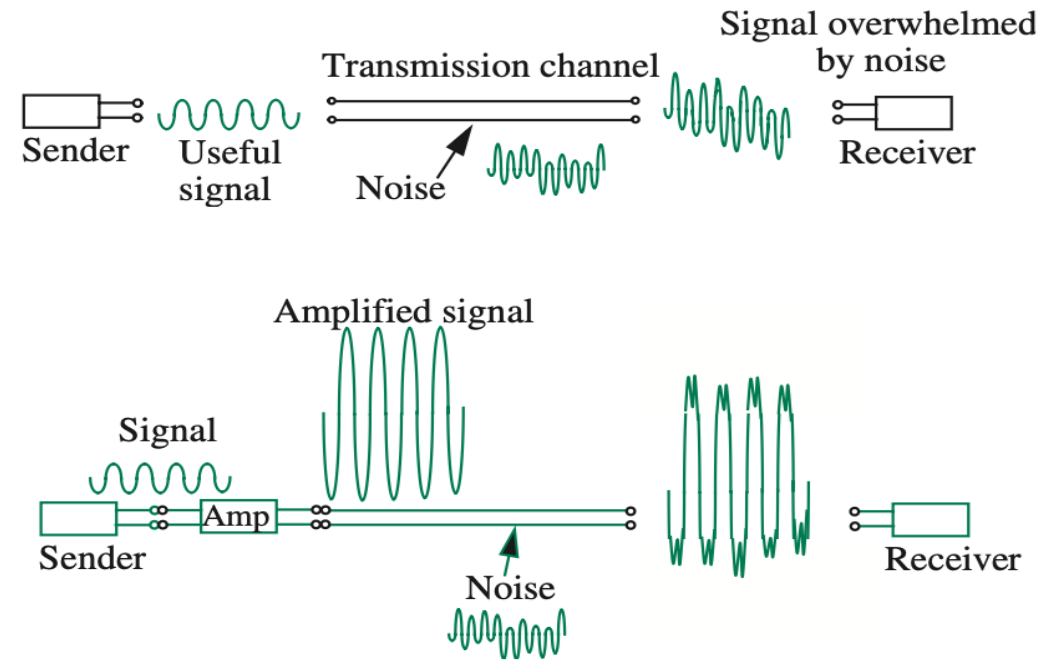
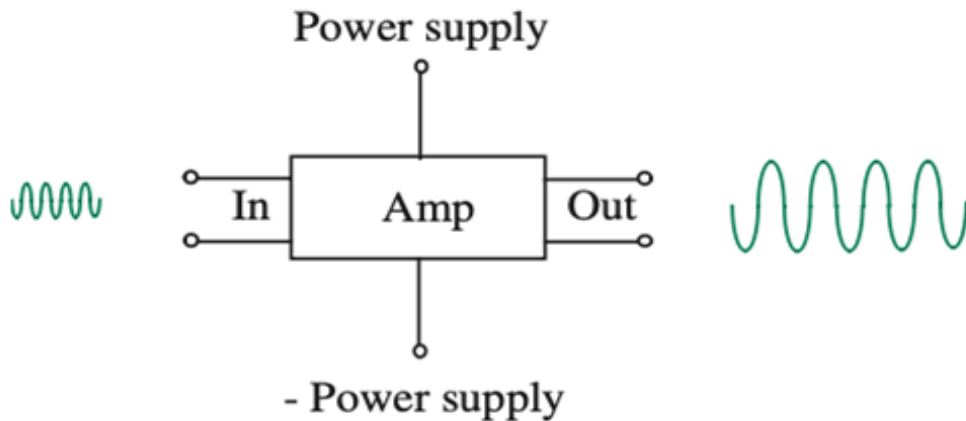
$$\text{Ans: } V = IR \Rightarrow V = \left(\frac{K}{V}\right) R$$

$$\Rightarrow V^2 = KR \Rightarrow V = \sqrt{KR}$$



Amplifier

- Linear amplifier $\Rightarrow OUT = k \times IN$. k is called the **gain** of the amplifier
- OUT/IN can be voltage or current
- Amplifier must provide power gain ($P_{OUT} > P_{IN}$), hence need **power supply**
- Usage: communication



Transfer Characteristics of Amplifier

- ❑ It's the relationship between **Input voltage (V_{in})** and **Output Voltage (V_{out})** of an amplifier.
- ❑ When $V_{in} \rightarrow$ x-axis and $V_{out} \rightarrow$ y-axis, the graph is **Voltage Transfer Characteristics (VTC)**.
- ❑ For a linear device, the VTC plot is $y = mx$ or $V_{out} = \text{Gain} \times V_{in}$, where slope (m) of the graph indicates the gain of the amplifier, $\text{Gain} = \frac{V_{out}}{V_{in}}$.

- When, $V_{out} > V_{in}$,

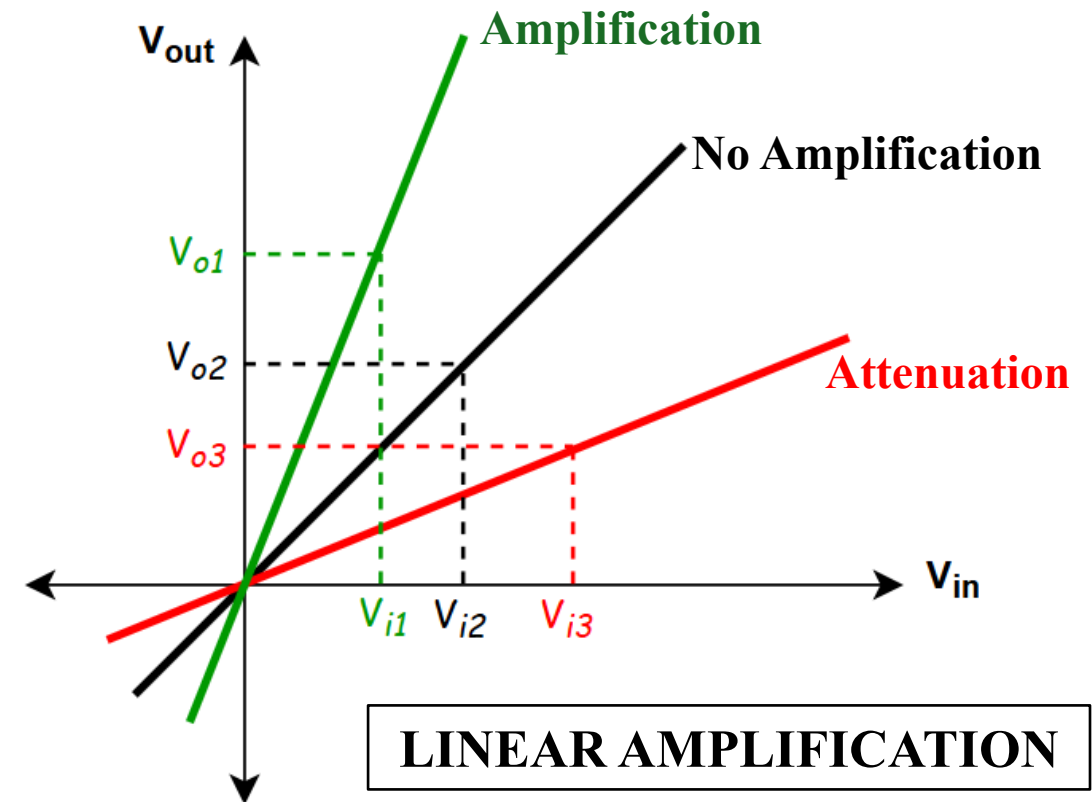
Amplification: $\text{Gain} = \frac{V_{o1}}{V_{i1}} > 1$

- When, $V_{out} = V_{in}$,

No Amplification: $\text{Gain} = \frac{V_{o2}}{V_{i2}} = 1$

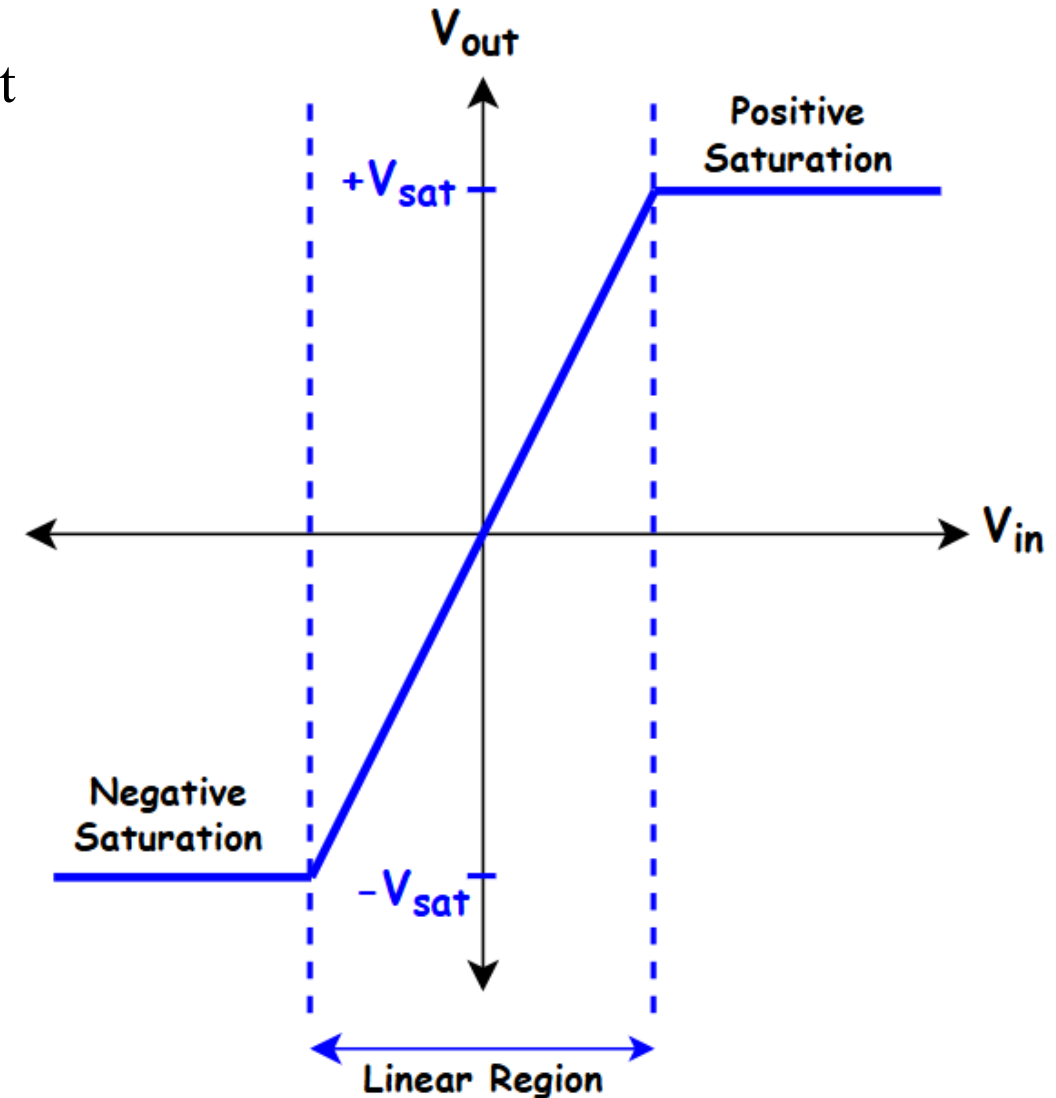
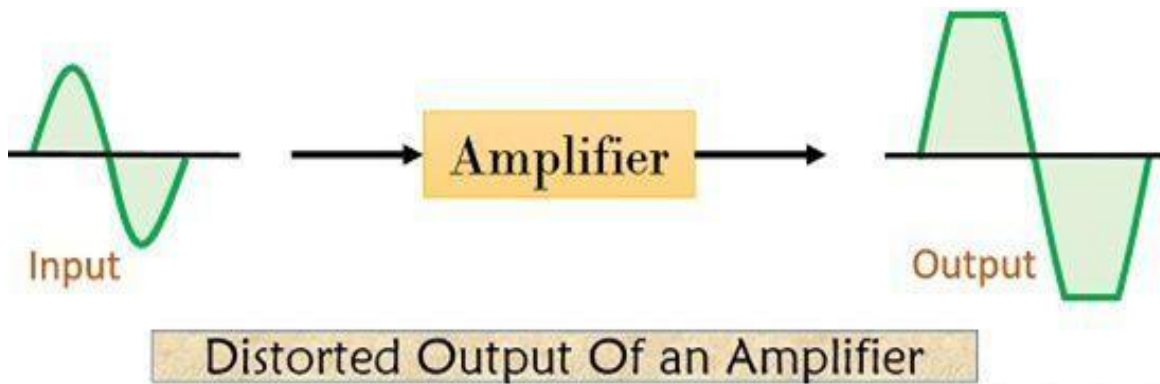
- When, $V_{out} < V_{in}$,

Attenuation: $\text{Gain} = \frac{V_{o3}}{V_{i3}} < 1$



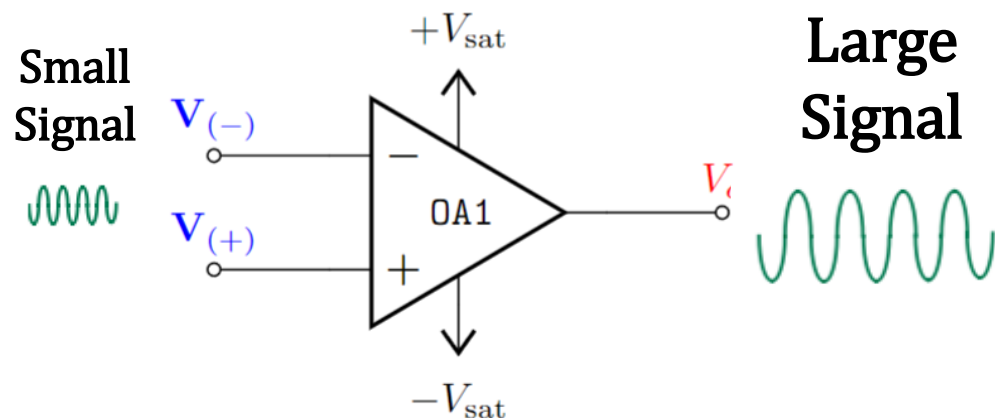
Transfer Characteristics of Amplifier

- ❑ Practically, Power Supply is limited, and output cannot be greater than the Supply Power.
- ❑ Output gets **saturated** at the Power Supply Levels.
- ❑ Input must be within **Valid Input Range**, for which output is within the power supply limit. **Otherwise, Output will be distorted.**
- ❑ Overall VTC of Amplifiers is **non-linear**, linear amplification occurs only within the linear region.



»» Operational Amplifier (Op-Amp)

- Operational Amplifier, or **Op-Amp** for short, is a versatile and powerful integrated circuit that is widely used in a variety of electronic applications.
- An Op-Amp is designed so that it performs some **mathematical operations** when external components, such as resistors and capacitors, are connected to its terminals.
- It is an electronic device consisting of a complex arrangement of resistors, transistors, capacitors, and diodes. A full discussion of what is inside the op amp is beyond the scope of this course. For now, it will suffice to treat the op amp as a circuit building block and simply study what takes place at its terminals.



•Operational:

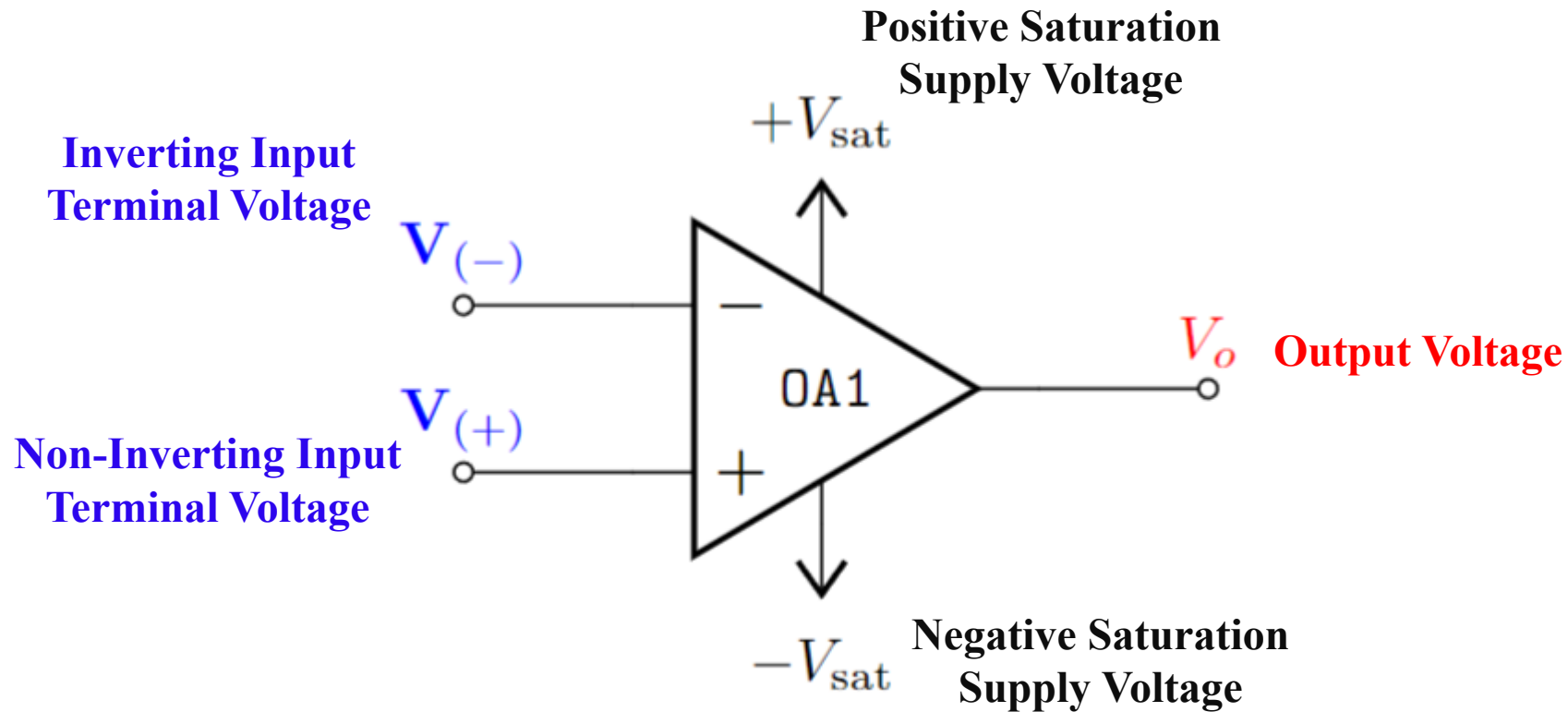
Mathematical Operations: Add, Subtract, Differentiation, Integration, Logarithm etc.

•Amplifier:

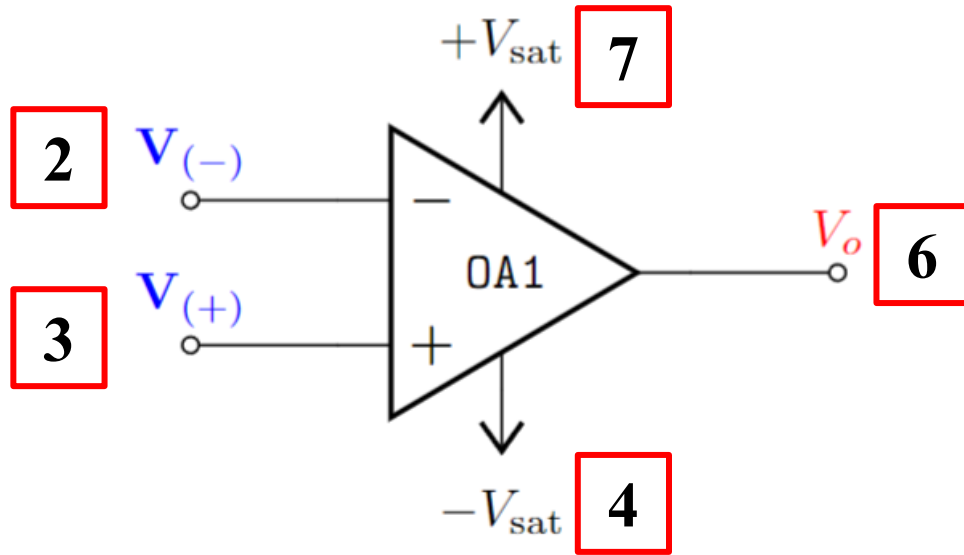
Amplifies input signal/voltage.

»» Op-Amp: Circuit Symbol and Terminals

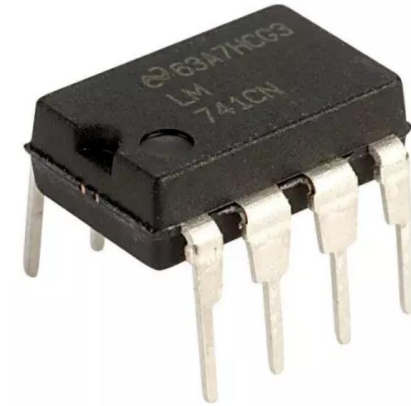
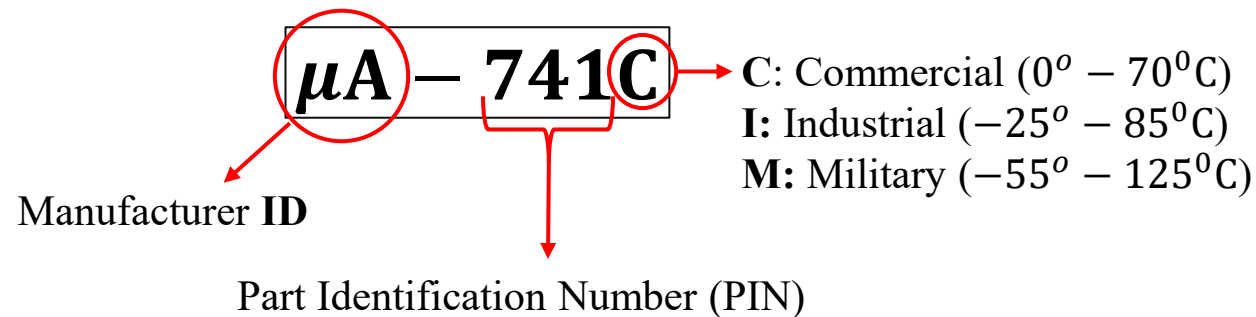
Total **5** terminals (nodes) with Op-Amp: **Two Input** Terminals [$V_{(+)}$ and $V_{(-)}$], **Two Power Supply** Terminals [$+V_{sat}$ and $-V_{sat}$], and **One Output** Terminal (V_o).



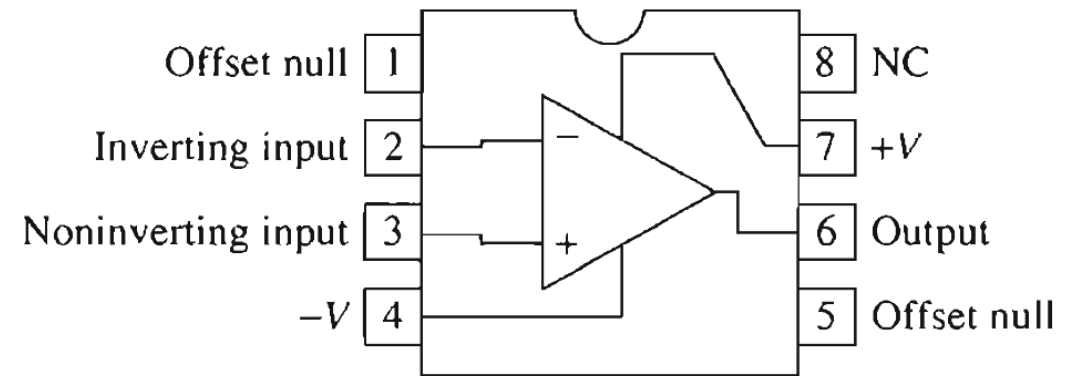
Op-Amp: Physical Entity



Circuit symbol for the general-purpose op amp.
Pin numbering is that for an **8-pin mini-DIP package**



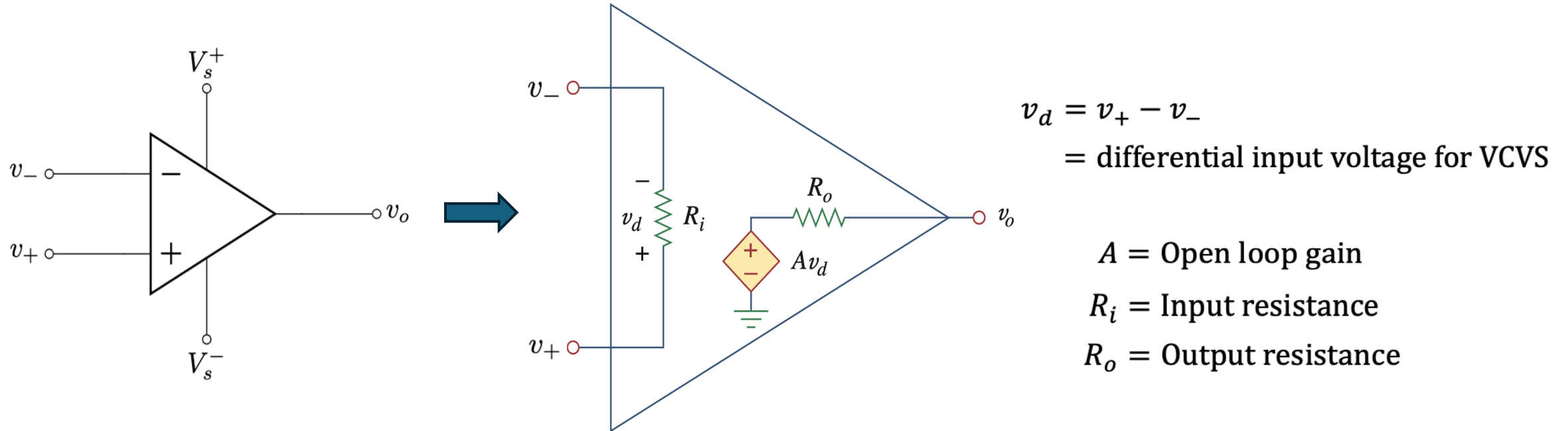
NC – No Connection



Op-Amp – PIN Diagram

Op-Amp: Introduction

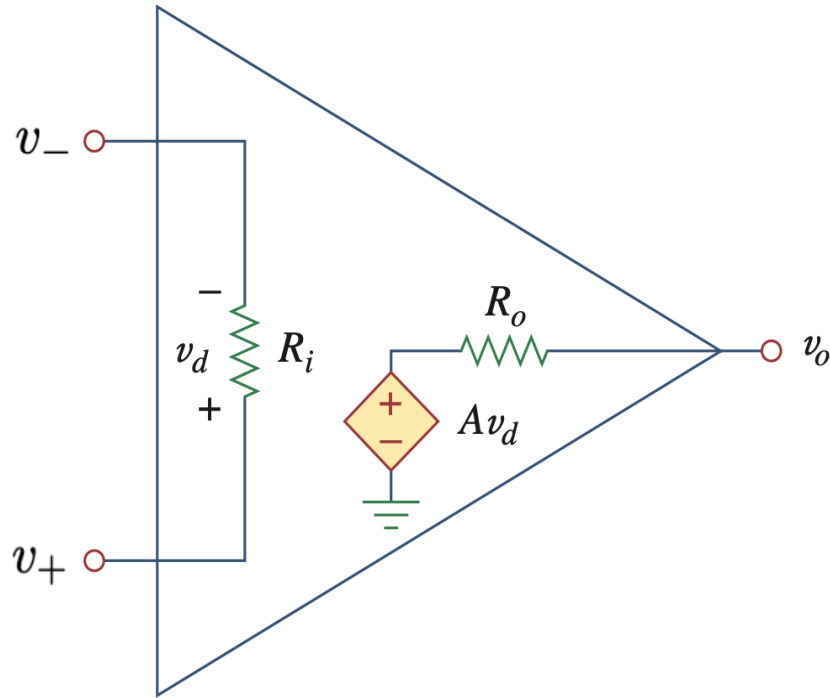
The internal circuit can be modeled using a **voltage controlled voltage source** (VCVS)! (actual circuit is complicated).



The op-amp senses the difference between the two inputs (V_d), multiplies by gain A , and causes the resulting voltage to appear at the output node (V_o). Hence, Op-amp is a **Difference Amplifier**, it amplifies the voltage difference between two input terminals.

$$V_o = AV_d = A(V_+ - V_-); \text{ not } A(V_- - V_+)$$

Op-Amp: Introduction



Equivalent Circuit of a **Real Op-Amp**

A = Open loop gain R_i = Input resistance

R_o = Output resistance

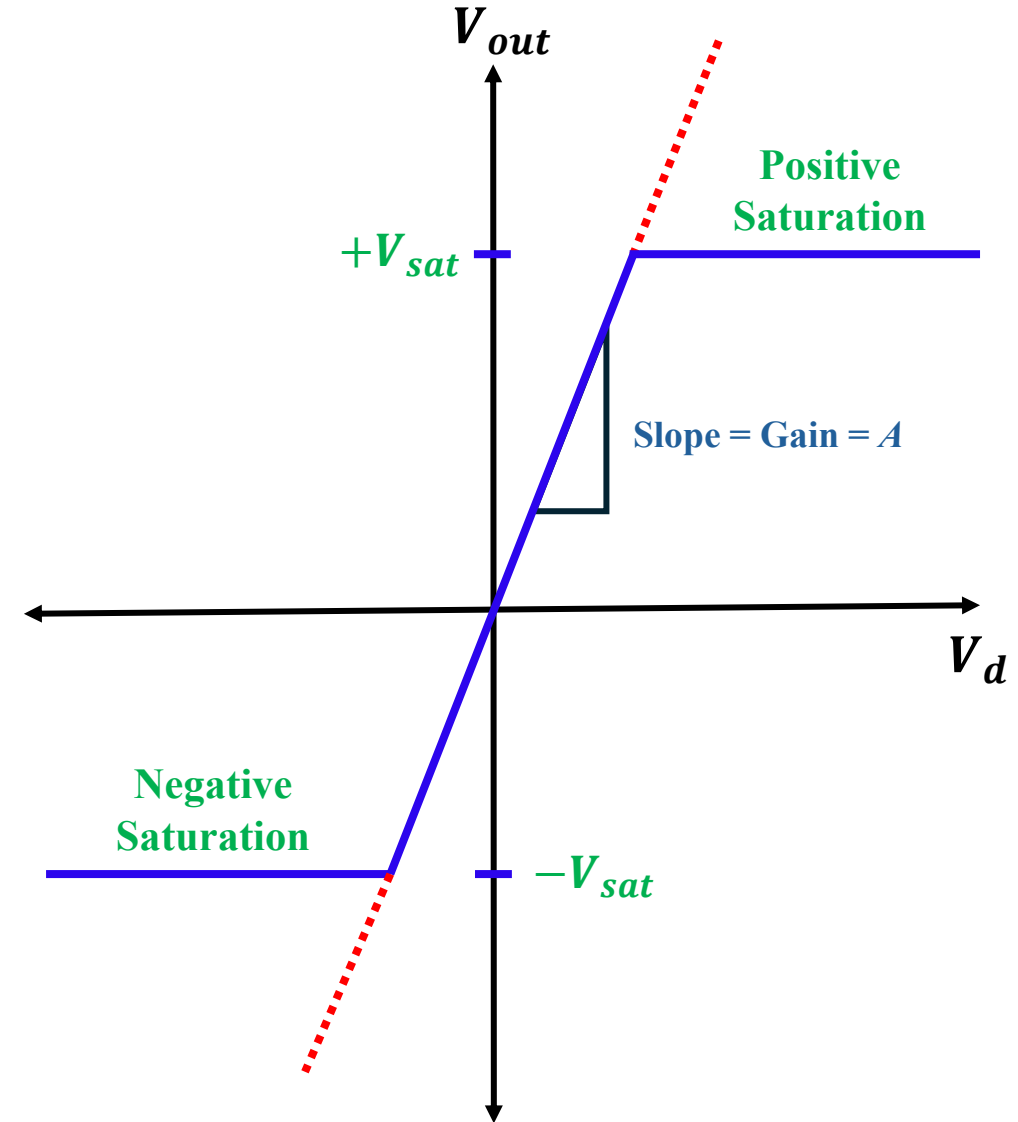
Parameter	Typical Range in Practical Op-Amps
A	$10^5 - 10^6$
R_i	$10^6\Omega - 10^{12}\Omega$
R_o	$10\Omega - 100\Omega$

- Op-Amps have a very **high Open Loop gain**; a small input voltage difference (V_d) would cause a very high output voltage (V_o) as:

$$V_o = AV_d = A(V_+ - V_-)$$
- However, the output voltage (V_o) cannot increase infinitely. It is limited by the Power Supply to the Op-Amp. This is called Saturation of Op-Amp.**

Op-Amp: Introduction

- Two Power Supply Terminals in Op-Amp: $+V_{sat}$ and $-V_{sat}$.
- Output cannot be greater than $+V_{sat}$. When $V_o > +V_{sat}$ for a certain input voltage difference, Output would be fixed at $+V_{sat}$. This is called **Positive Saturation**.
- Output cannot be less than $-V_{sat}$. When $V_o < -V_{sat}$ for a certain input voltage difference, Output would be fixed at $-V_{sat}$. This is called **Negative Saturation**.
- So, irrespective of the Input Voltage Difference, Output is always within the Power Supply Limit.
$$-V_{sat} \leq V_o \leq +V_{sat}$$
- The resulting VTC is non-linear due to saturation at the Power Supply levels, with $V_d = V_+ - V_-$ in the x-axis and V_o in the y-axis.



Op-Amp: Introduction

- Amplification is linear when V_d remains within the **Valid Input Range**. This is the **Linear Region** for Op-Amp.

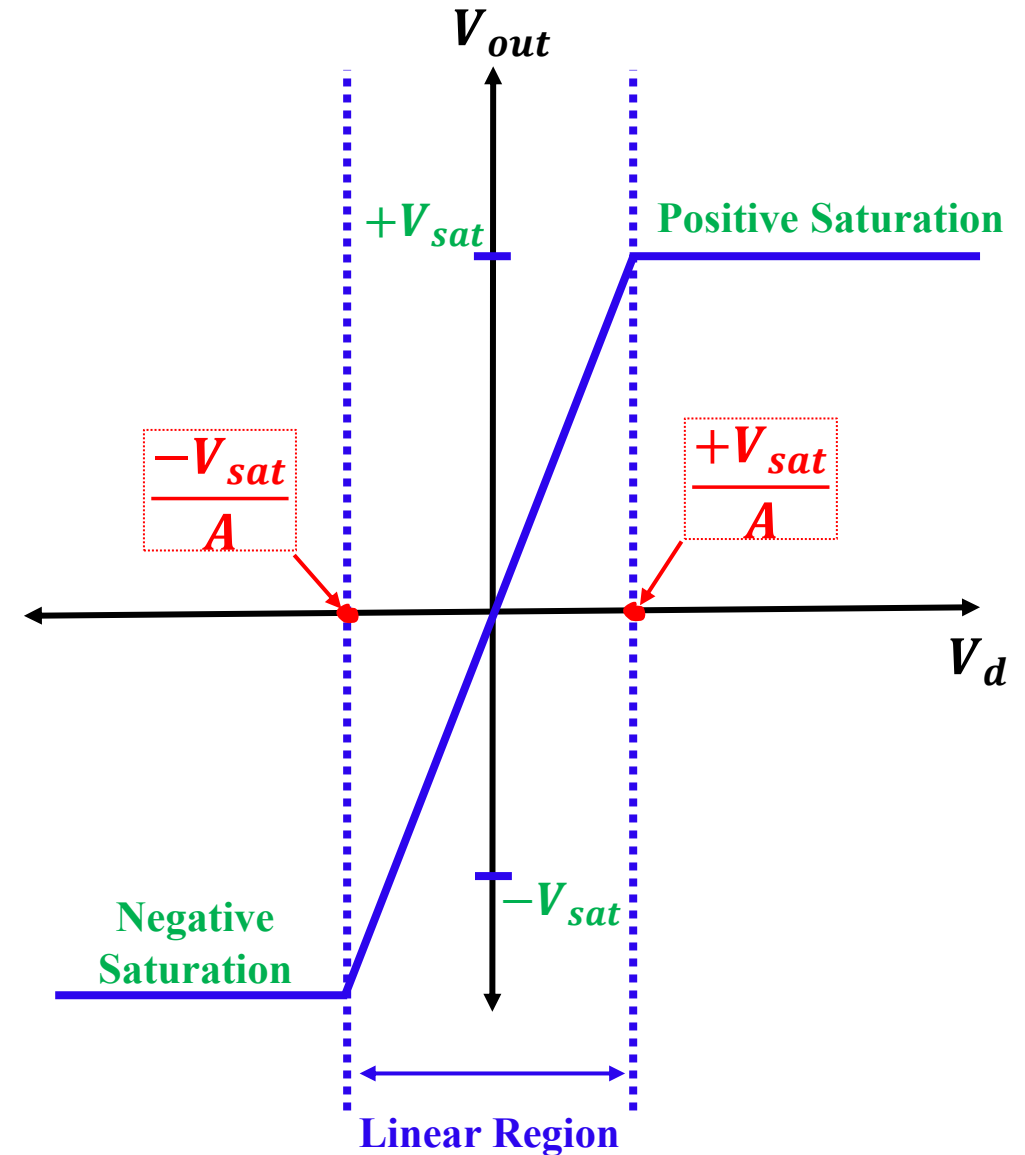
- Maximum Input voltage Difference** for **Linear** Amplification

$$= \frac{\text{Maximum Output Voltage}}{\text{Gain}} = \frac{+V_{sat}}{A}$$

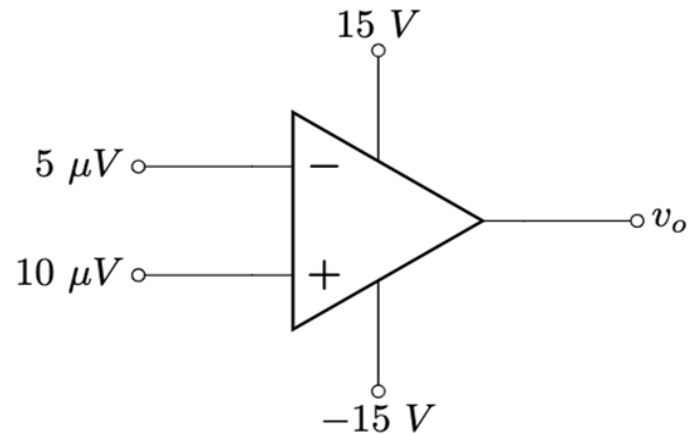
- Minimum Input voltage Difference** for **Linear** Amplification

$$= \frac{\text{Minimum Output Voltage}}{\text{Gain}} = \frac{-V_{sat}}{A}$$

- Valid Input Range:** $\frac{+V_{sat}}{A} \leq V_d \leq \frac{-V_{sat}}{A}$
- When V_d is beyond the Valid Input Range, output gets saturated at $+V_{sat}$ or $-V_{sat}$.



Examples



$$A = 2 \times 10^5$$

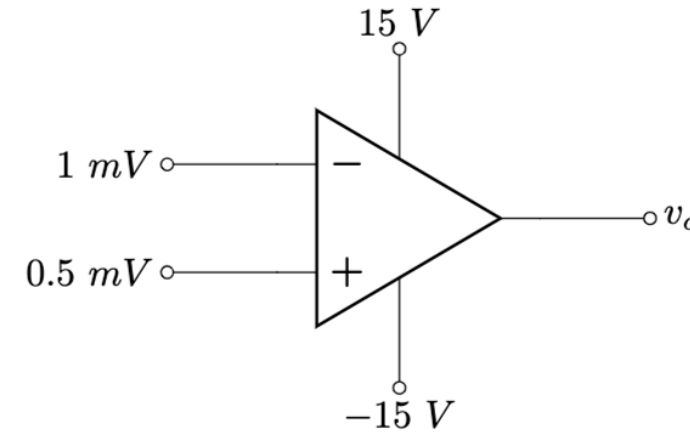
Solution:

$$\begin{aligned} V_d &= V_+ - V_- \\ &= (10 - 5)\mu V = 5\mu V \\ V_o &= AV_d = 2 \times 10^5 \times 5 \times 10^{-6} \\ &= 1V \end{aligned}$$

Since, output is within the saturation limits

$$-15 \leq V_o = 1V \leq 15$$

Hence, Answer is **1V**.



$$A = 2 \times 10^5$$

Solution:

$$\begin{aligned} V_d &= V_+ - V_- \\ &= (0.5 - 1)mV = -0.5mV \\ V_o &= AV_d = 2 \times 10^5 \times -0.5 \times 10^{-3} \\ &= -100V \end{aligned}$$

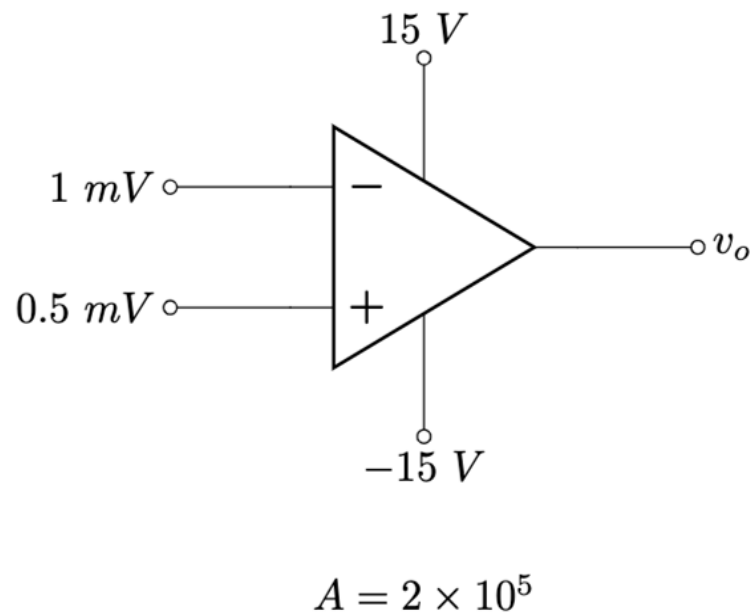
Since, output is less than $-V_{sat}$, that is

$$V_o = -100V < -V_{sat} (-15V)$$

Hence, Answer is **$V_o = -V_{sat} = -15V$** .

Examples

Calculate the Valid Input Range for this Op-Amp.



Solution: Maximum Input Voltage Difference

$$= \frac{+V_{sat}}{A} = \frac{15}{2 \times 10^5} = 75\mu V$$

Minimum Input Voltage Difference

$$= \frac{-V_{sat}}{A} = \frac{-15}{2 \times 10^5} = -75\mu V$$

So, Valid Input Range:

$$-75\mu V \leq V_d \leq 75\mu V$$

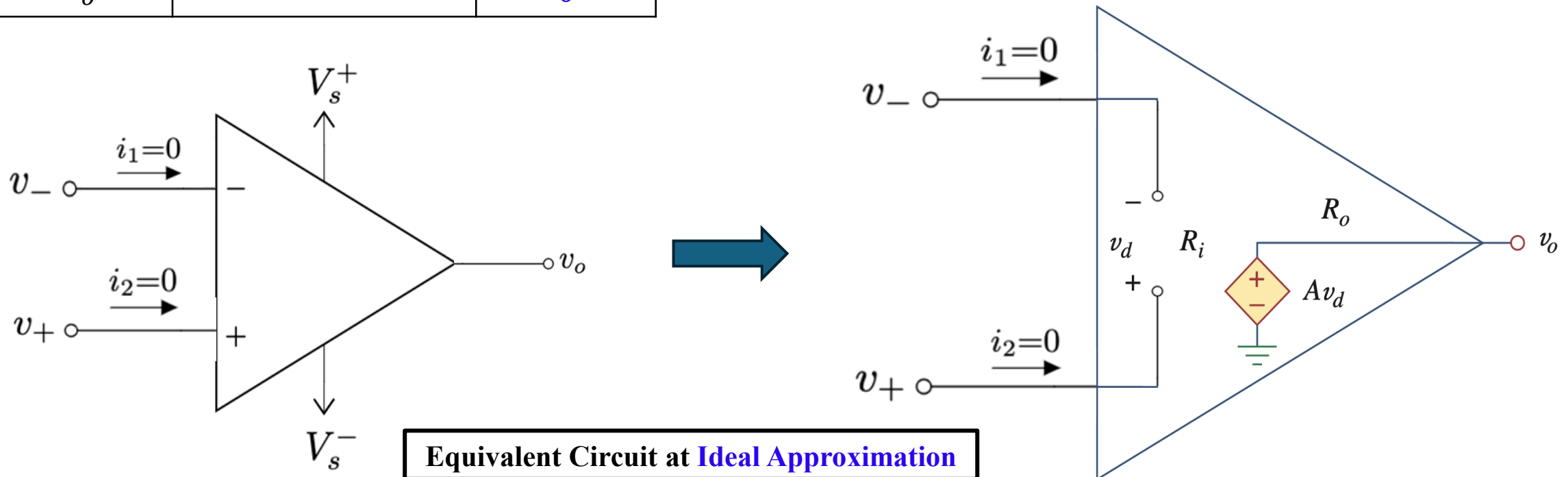
Note: While finding the valid input range, the values of input terminals V_+ and V_- don't matter.

➤➤➤ Ideal Op-Amp Approximation

To facilitate the understanding of Op-Amp Circuits, **we will approximate Ideal Op-Amp.**

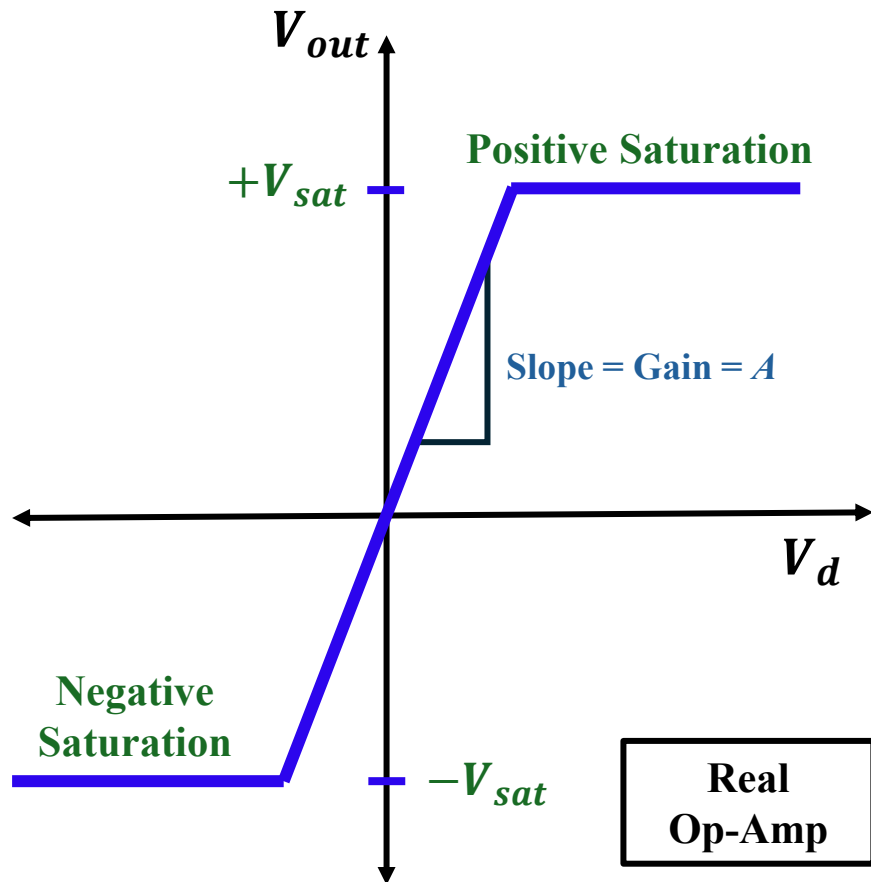
Parameter	Typical Range	Ideally
A	$10^5 - 10^6$	∞
R_i	$10^6\Omega - 10^{12}\Omega$	∞
R_o	$10\Omega - 100\Omega$	0

- R_i becomes **open-circuit.**
- R_o becomes **short-circuit.**
- **Since $R_i = \infty$, So, $i_- = i_+ = 0$**
- Circuit Solving becomes much easier.



➤➤➤ Ideal Op-Amp: VTC

Remember,
Slope of VTC = Gain of Op-Amp = A



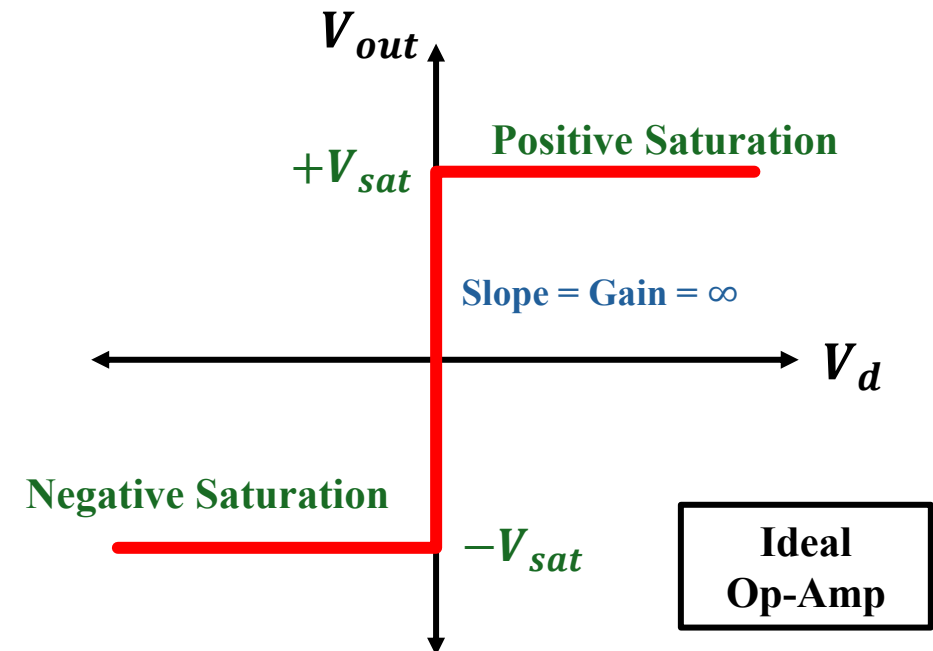
In ideal Op-Amp, $A = \infty$; which means slope of VTC = ∞
So, when $V_+ > V_-$ or $V_d = (V_+ - V_-) > 0$

$$V_{out} = AV_d = +V_{sat} \text{ (Positive Saturation)}$$

Similarly, for $V_+ < V_-$ or $V_d = (V_+ - V_-) < 0$

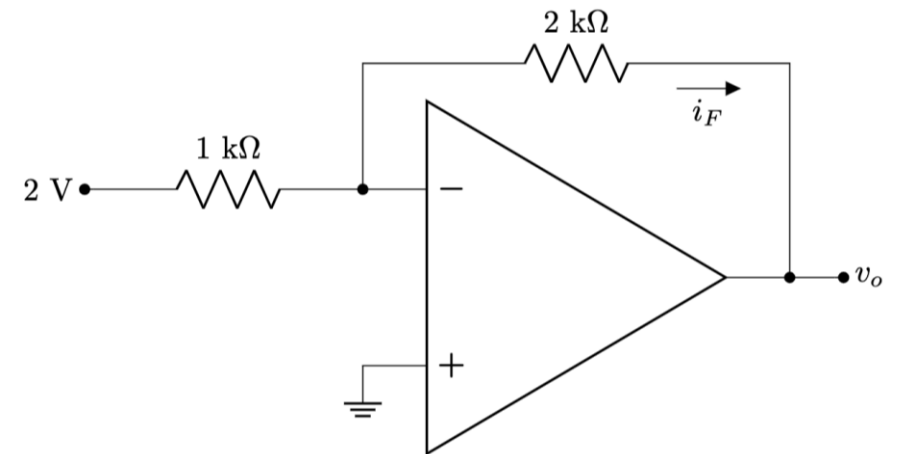
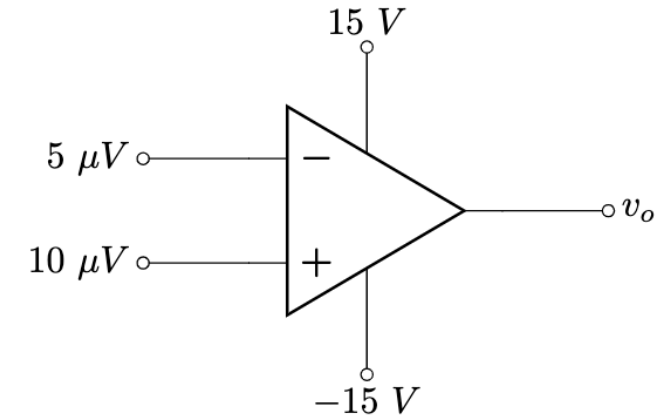
$$V_{out} = AV_d = -V_{sat} \text{ (Negative Saturation)}$$

No Linear Amplification, Output is either $+V_{sat}$ or $-V_{sat}$.



Op-Amp Configurations

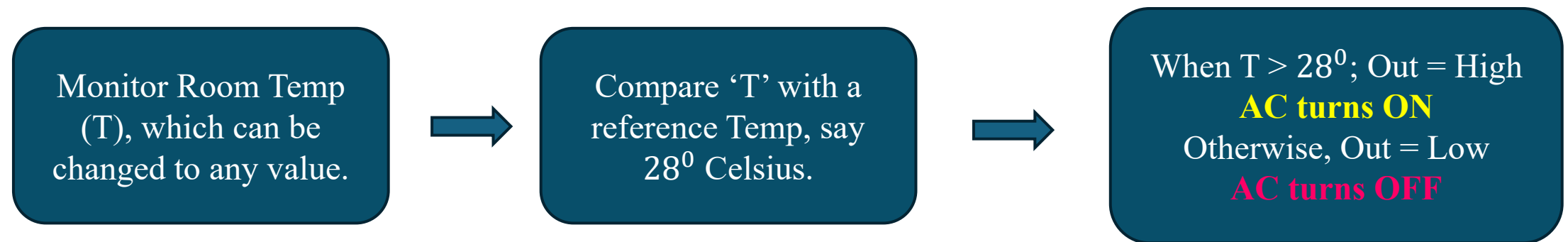
- 1. Open Loop Configuration:** No Physical Connection Between Output and Input terminals of Op-Amp. So far, we have discussed this configuration. Application in Open Loop: **Comparator**, which we will see next.
- 2. Closed Loop Configuration:** There is a Physical Connection (feedback) from the Output terminal to the Input terminal. We shall discuss Closed Loop Configuration in the next Slide. Application in Closed Loop: **Amplifier, Mathematical Operation, such as Addition, Subtraction, Integration, Differentiation, Logarithm, etc.**



»»» Comparator

An Op-Amp Comparator is an electronic device that **compares the two Input signals** and **produces a high or a low Output Signal** based on a predefined condition. It essentially acts as a decision-making circuit. We consider **Ideal Op-Amps in Open Loop Configuration** to implement comparators.

Consider a scenario where you need to design a system that continuously monitors Room Temperature and automatically turn on the AC when Temperature exceeds a certain limit. Basically, the system is **comparing the room temperature with a fixed reference temperature**. Based on the comparison, the system turns ON or OFF the AC. This problem can be solved by designing a suitable comparator.



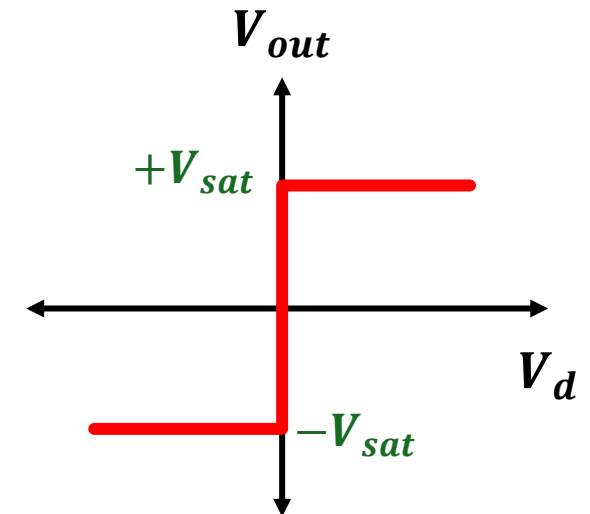
Comparator

Based on the previous example of AC,

- There is **a continuous signal** that is being monitored (the room temperature 'T'). It is the **Input signal (V_{in})** to the comparator. Practically, Input signal can have any value from $-\infty$ to $+\infty$.
- A **fixed reference value (V_{ref})** with which the Input signal is being compared (28° Celsius). This reference value ideally doesn't change while operation.
- Output can be either High Voltage V_H (AC ON) or Low Voltage V_L (AC OFF). **Output is discrete.**

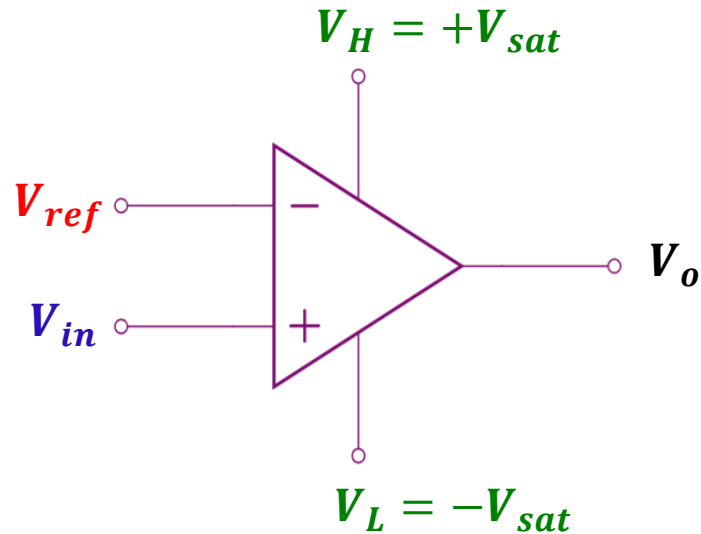
So, how to solve this problem using Ideal Op-Amp in Open Loop?

- One input terminal of Op-Amp receives the Input signal (V_{in}).
- Another input terminal holds the reference value (V_{ref}).
- For ideal Op-Amp in open loop, output is either $+V_{sat}$ or $-V_{sat}$. So, we can set desired $V_H = +V_{sat}$ and desired $V_L = -V_{sat}$.
- This is well understood from the VTC of ideal Op-Amp (open loop). **But for comparator, x-axis of VTC will be changed.**



Types of Comparator

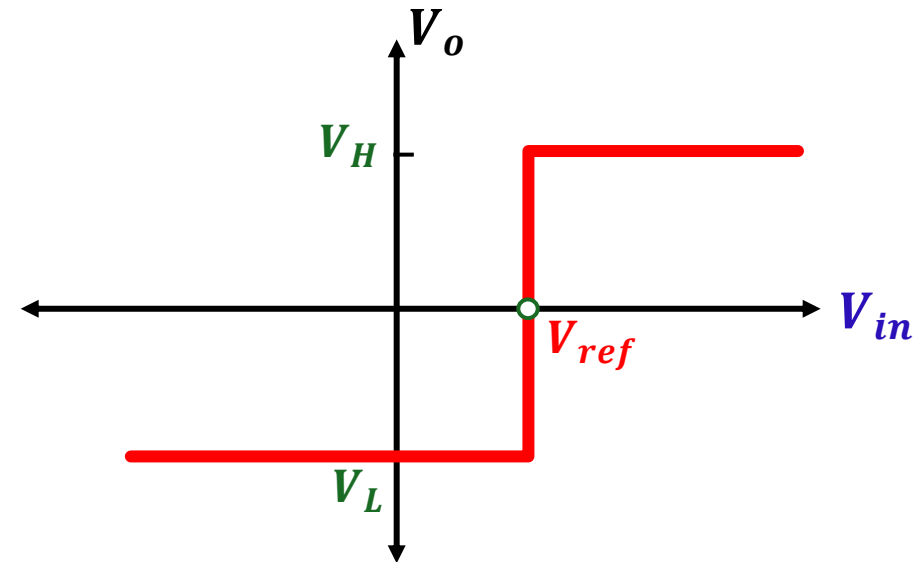
Non-Inverting Comparator:



- Set V_{in} in the V_+ input terminal.
- Set V_{ref} in the V_- input terminal.
- Set V_H in the $+V_{sat}$ supply terminal.
- Set V_L in the $-V_{sat}$ supply terminal.

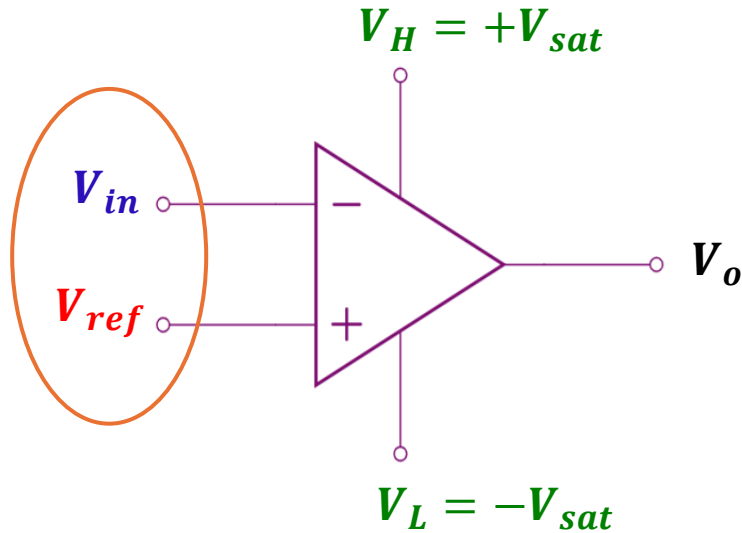
- When $V_{in} > V_{ref}$; $V_d = (V_{in} - V_{ref}) > 0$
 $V_{out} = V_H (+V_{sat})$
- When $V_{in} < V_{ref}$; $V_d = (V_{in} - V_{ref}) < 0$
 $V_{out} = V_L (-V_{sat})$

In the VTC, x-axis will be V_{in} now, not V_d . Because for comparator, we analyze the relationship between V_{in} from physical world with the V_o .



Types of Comparator

Inverting Comparator:

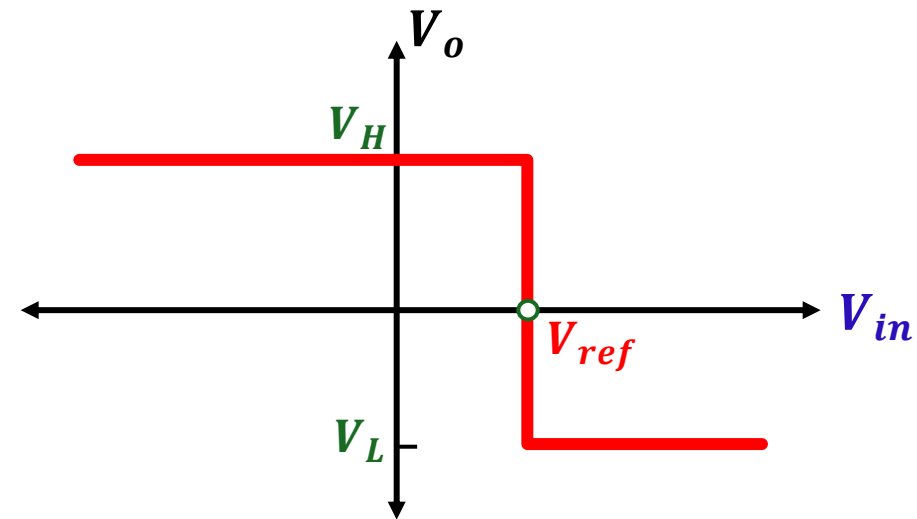


- Set V_{in} in the V_- input terminal.
- Set V_{ref} in the V_+ input terminal.
- Set V_H in the $+V_{sat}$ supply terminal.
- Set V_L in the $-V_{sat}$ supply terminal.

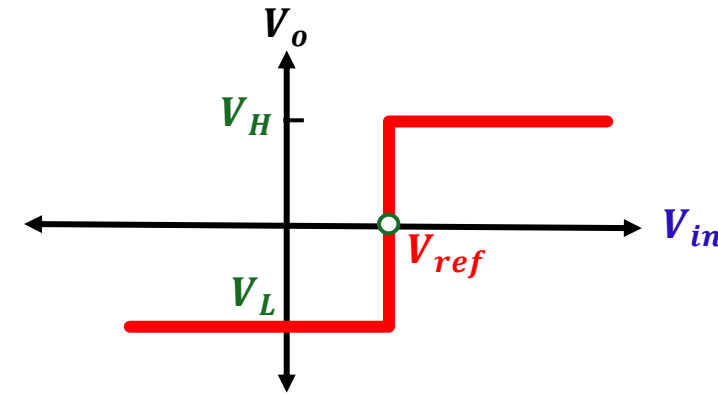
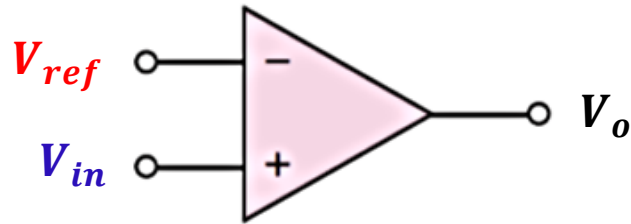
Notice, the Positions of V_{in} and V_{ref} are interchanged.

- When $V_{in} > V_{ref}$; $V_d = (V_{ref} - V_{in}) < 0$
 $V_{out} = V_L (-V_{sat})$
- When $V_{in} < V_{ref}$; $V_d = (V_{ref} - V_{in}) > 0$
 $V_{out} = V_H (+V_{sat})$

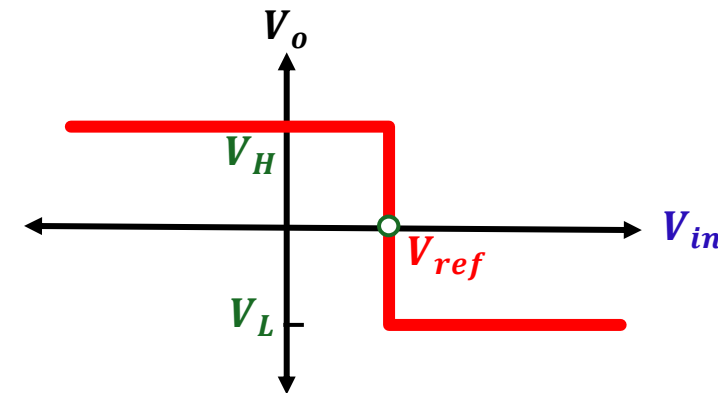
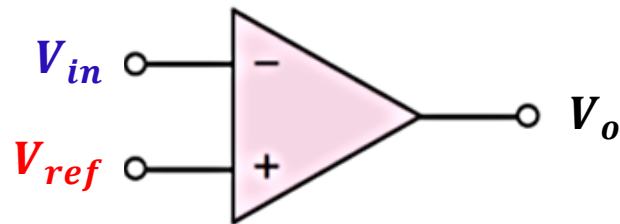
In the VTC, the relationship between V_{in} from physical world with the V_o will be as the following:



Identifying Comparator



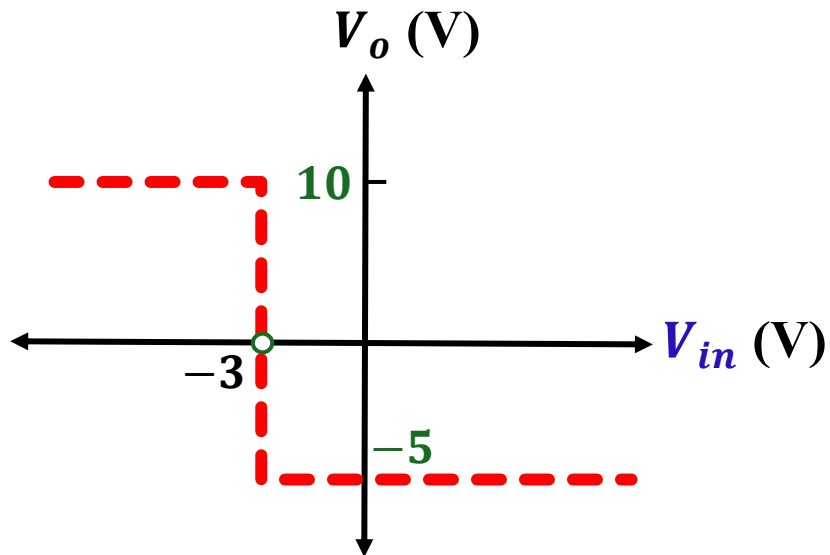
Non-Inverting Comparator: Input V_{in} is connected to the **Non-Inverting Input Terminal (V_+)**.
VTC looks like a **Step Up** in the Staircase.



Inverting Comparator: Input V_{in} is connected to the **Inverting Input Terminal (V_-)**.
VTC looks like a **Step Down** in the Staircase.

Example 01

Design a circuit using **Op-Amp** that has the voltage transfer characteristics as shown in the figure below. Here, V_o is the **Output voltage** and V_{in} is the **Input voltage**.



Solution:

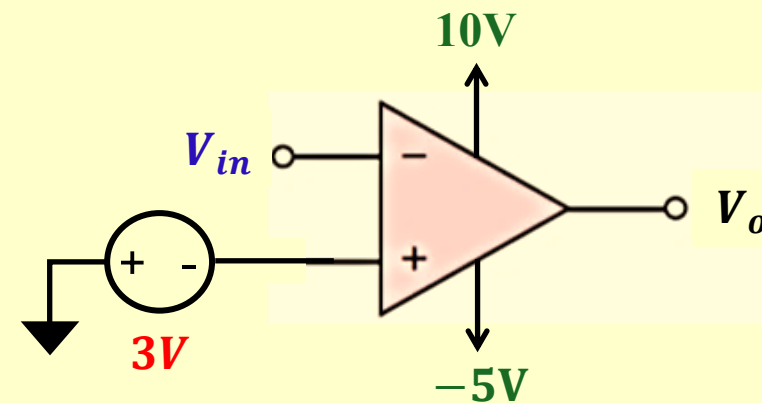
VTC looks like Step Down stair. So, Inverting Comparator.
Looking at the V_o axis,

$$\text{Max Output} = 10V = V_H$$

$$\text{Min Output} = -5V = V_L$$

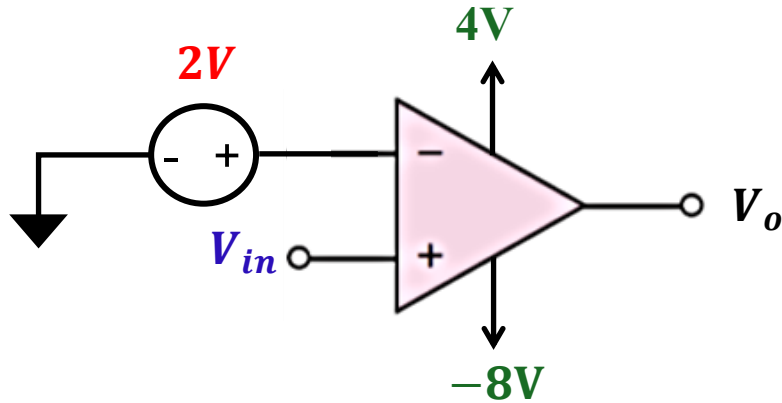
Looking at the V_{in} axis, output changes when $V_{in} = -3V$

$$\text{So, } V_{ref} = -3V$$



Example 02

Draw the voltage transfer characteristics (VTC) graph for the following Op-Amp Comparator Circuit. Here, V_o is the **Output voltage** and V_{in} is the **Input voltage**.



Solution:

V_{in} is connected to the V_+ terminal.

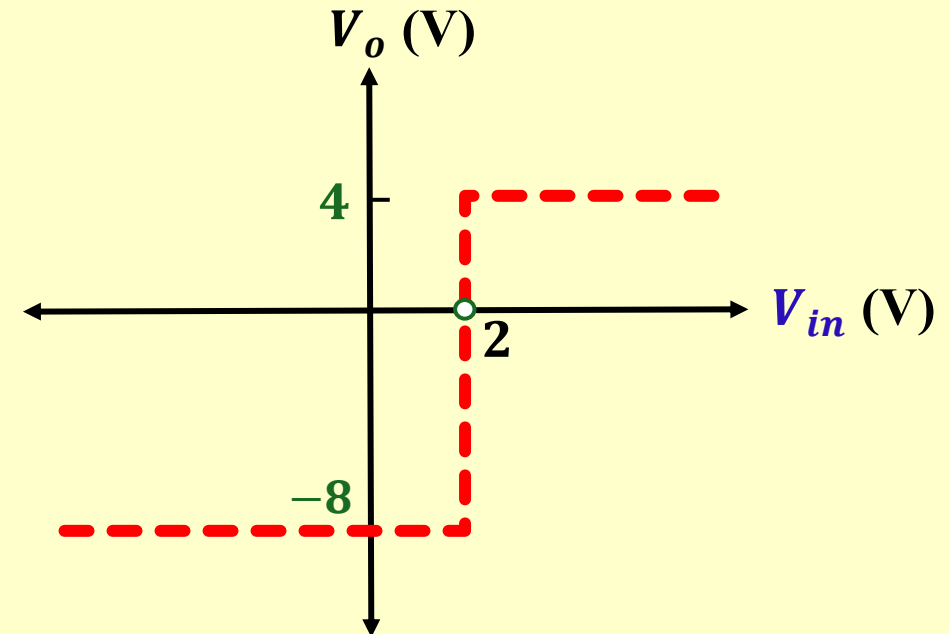
So, Non-Inverting Comparator.

Max Output = $V_H = 4V$

Min Output = $V_L = -8V$

Voltage at the V_- terminal = 2V, which is a fixed value.

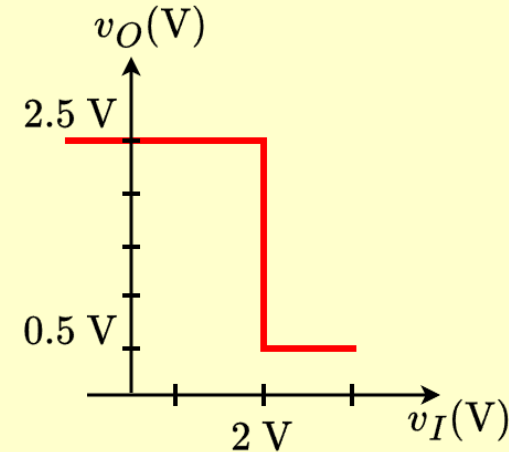
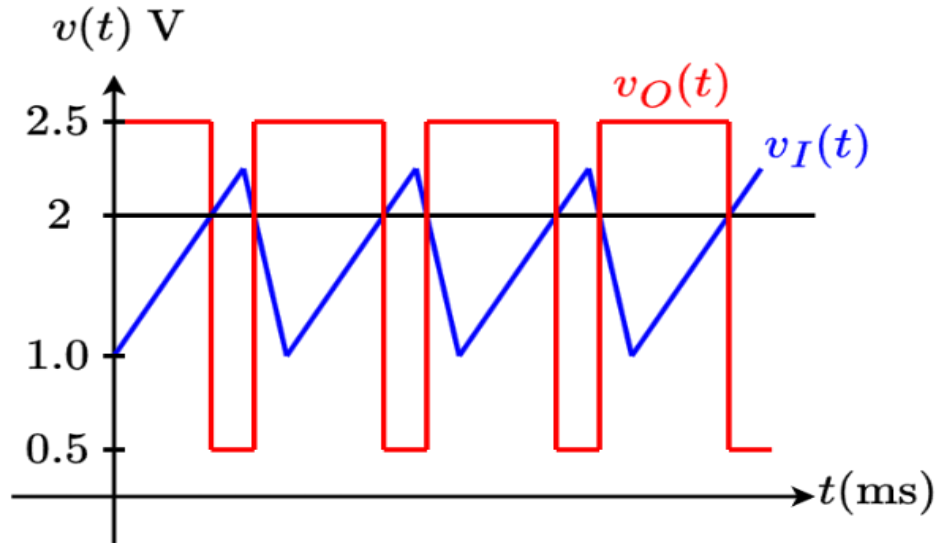
So, $V_{ref} = 2V$



The resulting VTC looks like a Step-Up stair.
 V_o shifts from -8V to 4V when V_{in} crosses 2V.

Example 03

Draw the voltage transfer characteristic (VTC) curve from the adjacent waveform graph. Also draw the **Op-Amp Circuit** that would give rise to such a VTC.



The resulting VTC looks like a Step-Down Staircase.

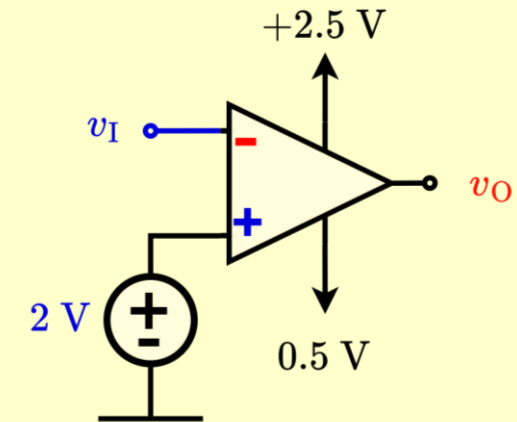
Inverting Comparator.

v_O changes when v_I crosses 2V. Hence, $V_{ref} = 2$ V.

Solution:

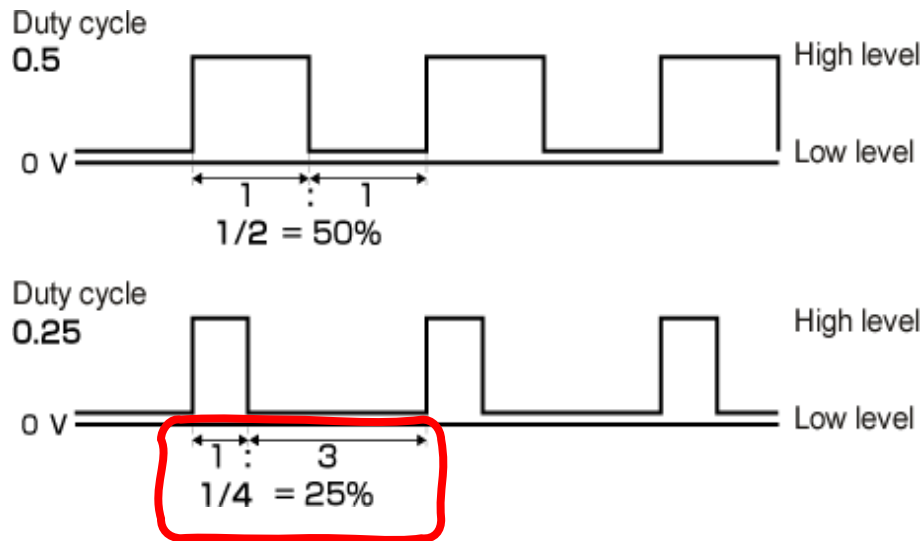
When, $v_I < 2$ V $\Rightarrow v_O = 2.5$ V $\rightarrow$ Positive Saturation

When, $v_I > 2$ V $\Rightarrow v_O = 0.5$ V $\rightarrow$ Negative Saturation

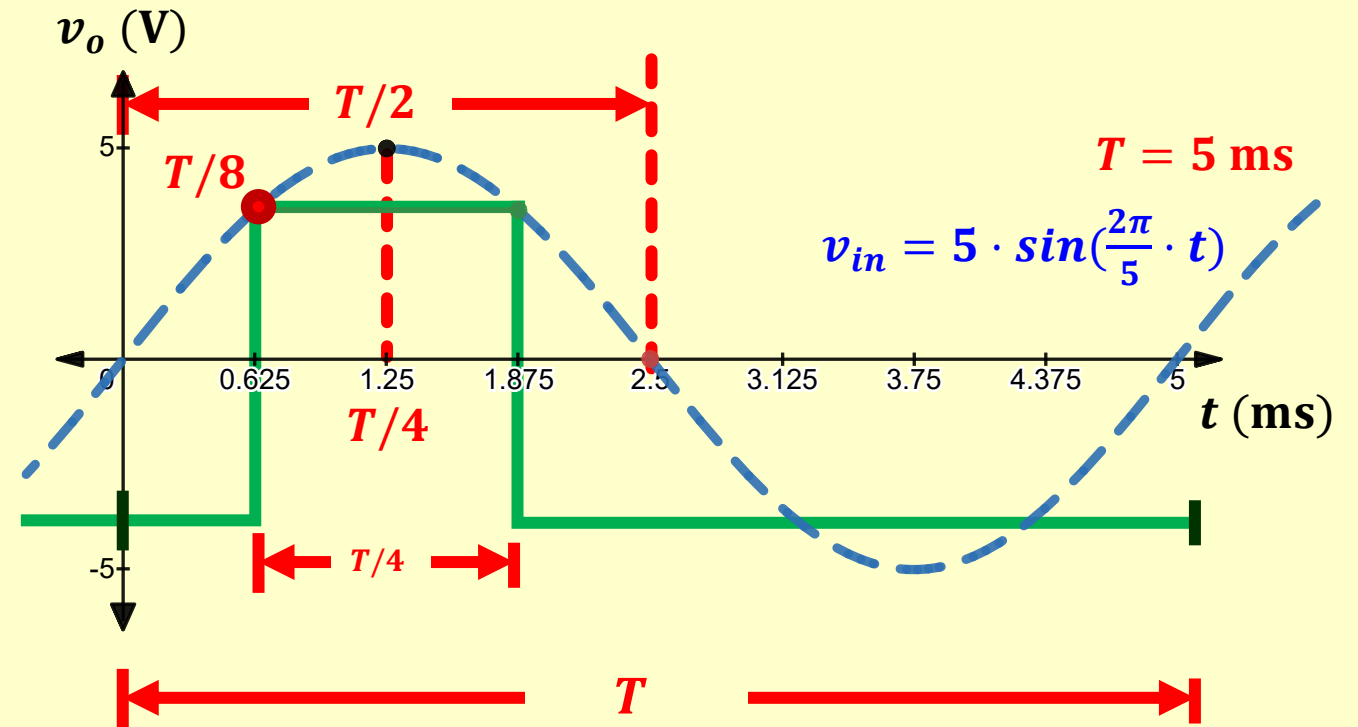


Example 04

Design an op-amp circuit to transform the sinusoidal voltage, $v_{in} = 5 \cdot \sin(\frac{2\pi}{5} \cdot t)$ (t is in units of ms , and time-period T is $5\ ms$), to a square wave with a duty cycle of **25%**.



Solution:



Example 04

Solution:

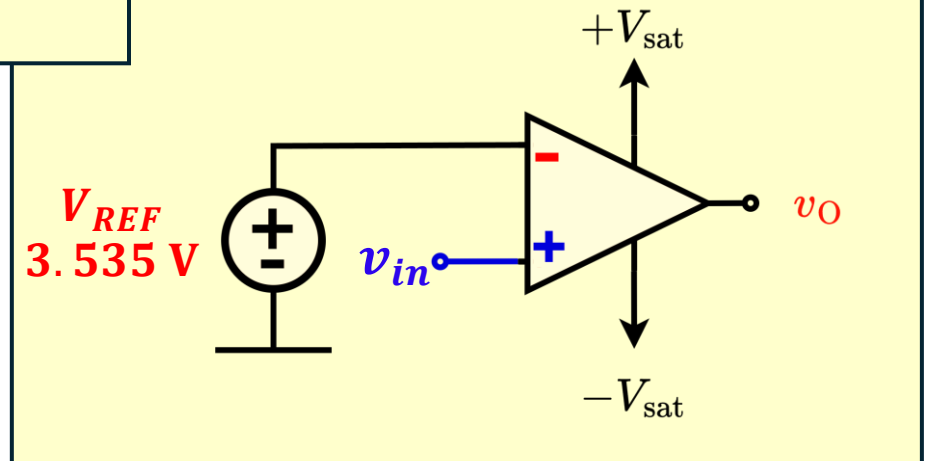
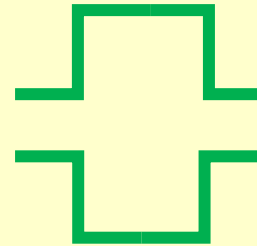
The time point when Output shifts from Low Level to High Level, $t = \frac{T}{8}$

$$\text{So, } V_{REF} = 5 \sin\left(\frac{2\pi}{5} \cdot \frac{T}{8}\right) = 5 \sin\left(\frac{2\pi}{5} \cdot \frac{5}{8}\right) = 3.535V$$

When $v_{in} \geq 3.535 \text{ V} : v_o \rightarrow \text{Positive Saturation}$

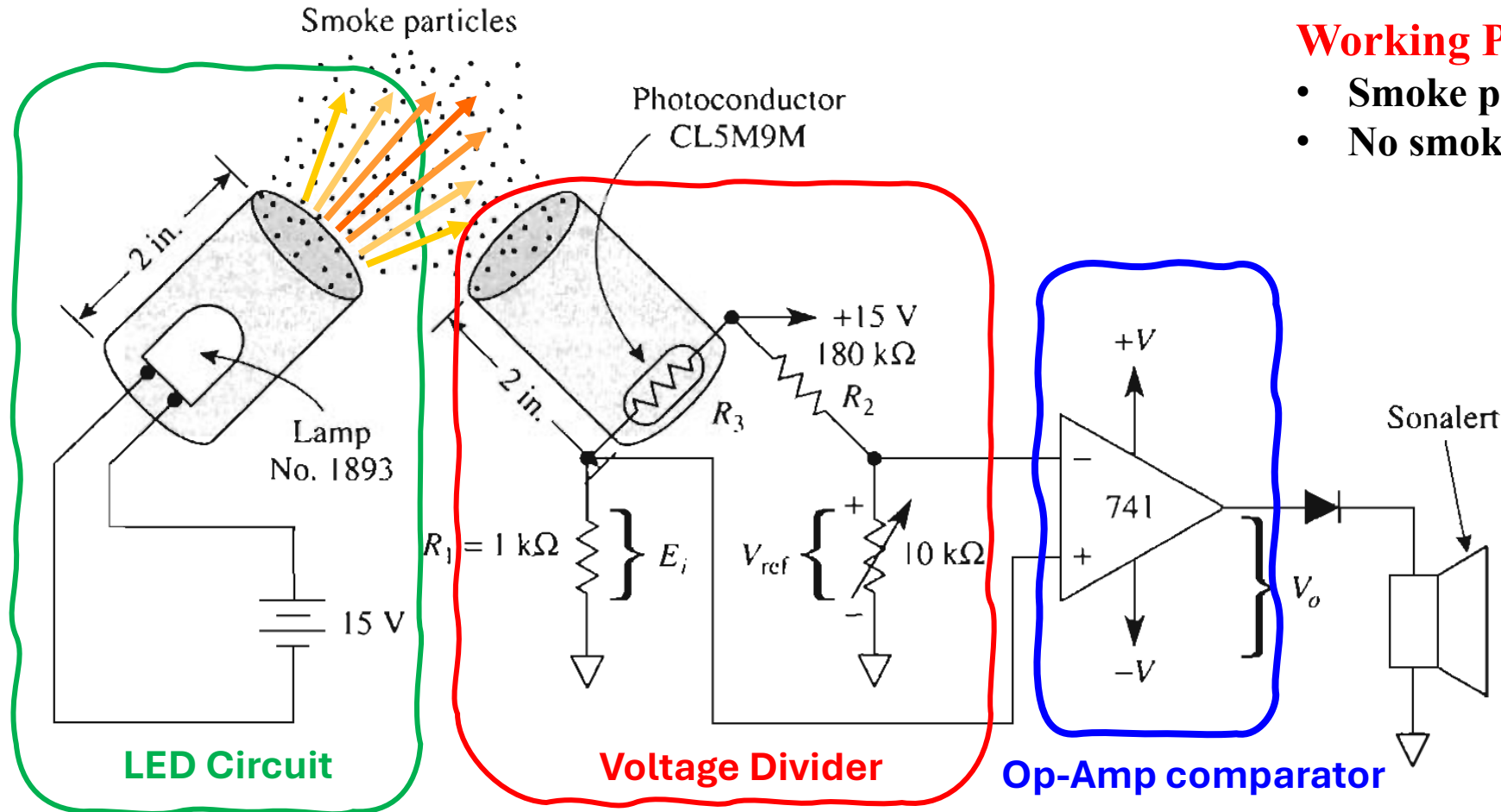
When $v_{in} \leq 3.535 \text{ V} : v_o \rightarrow \text{Negative Saturation}$

NON-INVERTING CONFIGURATION



Example 05: Smoke Detector

With no smoke or dust present the 10-k Ω sensitivity control is adjusted until the alarm stops. Light reflected off any smoke or dust particles causes the alarm to sound.



Working Principle:

- **Smoke present** → Alarm rings
- **No smoke present** → No Alarm rings.

Since the **alarm** is an electrically driven device, we need some method to convert smoke levels to proportionate voltage.

Smoke levels → Voltage

Example 05: Smoke Detector

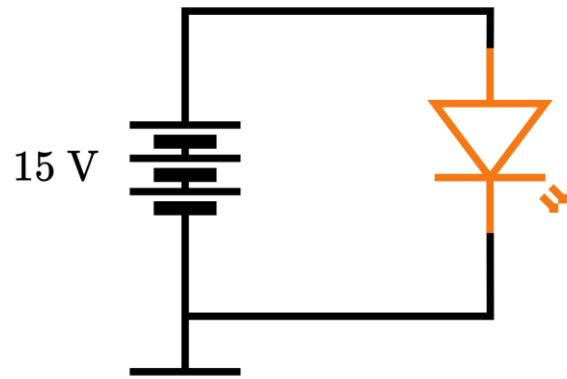
Light from the LED is reflected off the smoke particles on to the photoconductor –
Light Dependent Resistor (LDR).

No Smoke

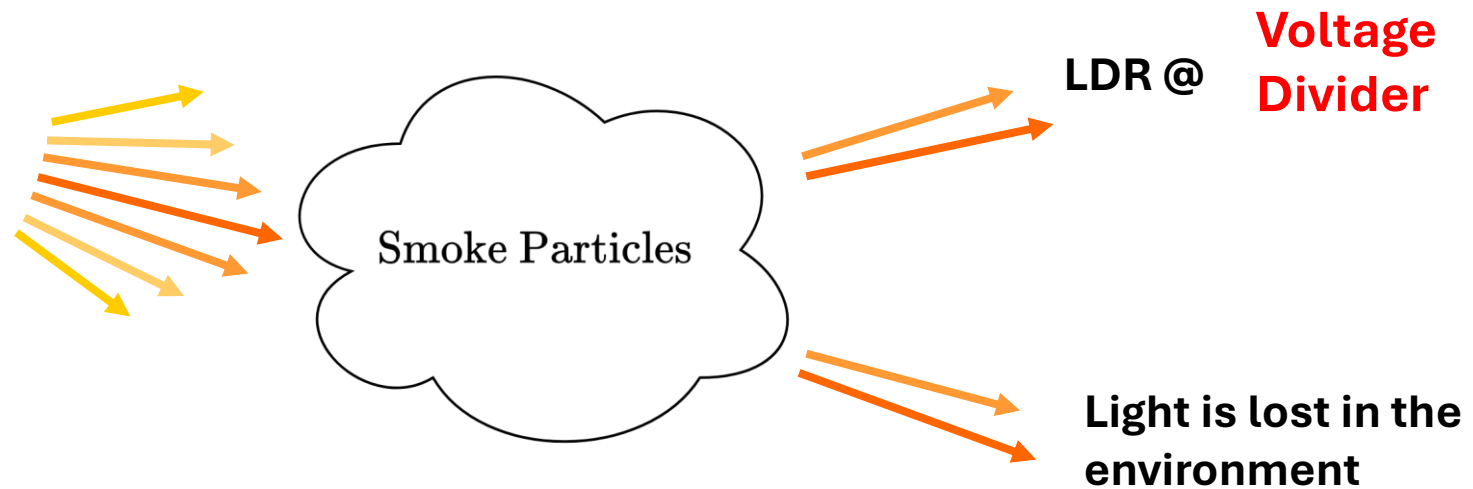
- No scattering
- Low intensity at LDR (Dark - $I \downarrow$)

Smoke Present

- High scattering from smoke particles
- High intensity at LDR (Light - $I \uparrow$)



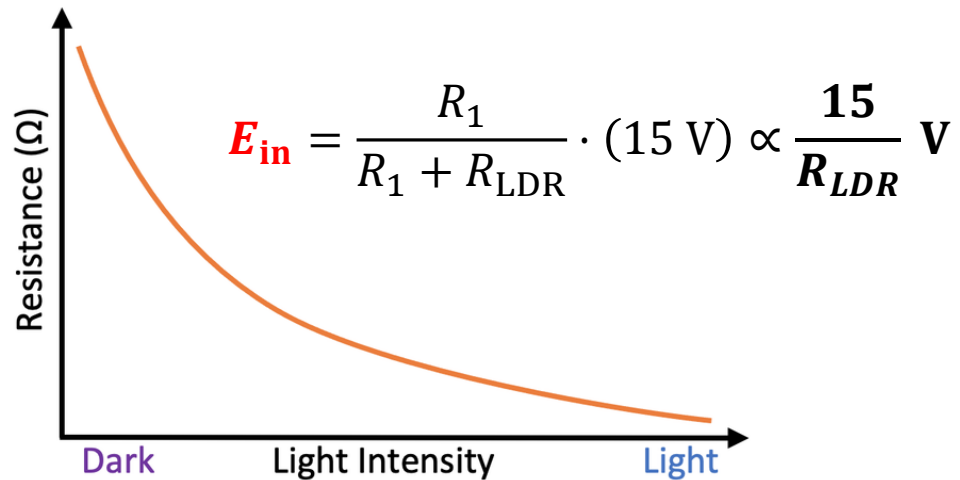
LED Circuit



Example 05: Smoke Detector

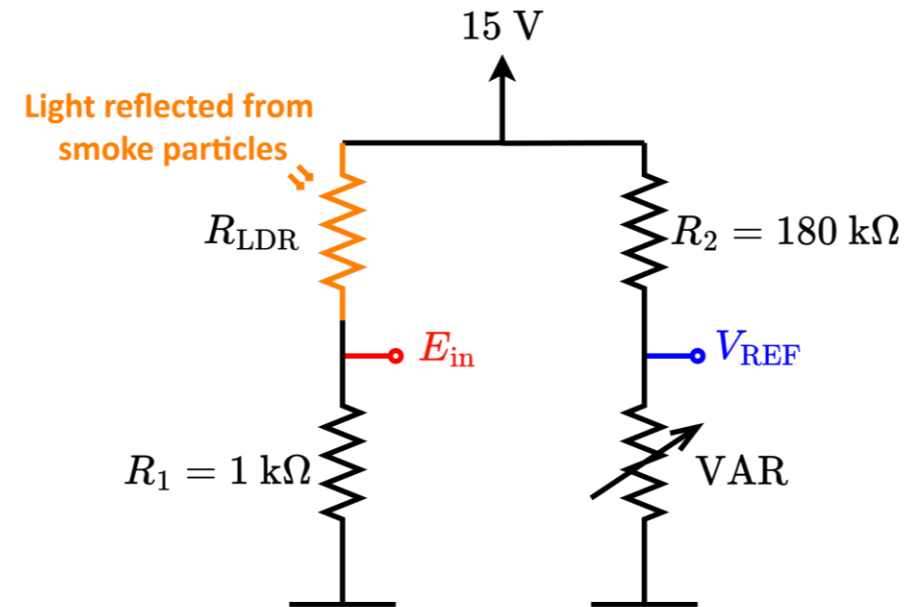
No Smoke

- No scattering
- Low intensity at LDR (Dark - $I \downarrow$)
- High resistance across LDR ($R_{LDR} \uparrow$)
- Decrease in E_{in}



Smoke Present

- High scattering from smoke particles
- High intensity at LDR (Light - $I \uparrow$)
- Low resistance across LDR ($R_{LDR} \downarrow$)
- Increase in E_{in}



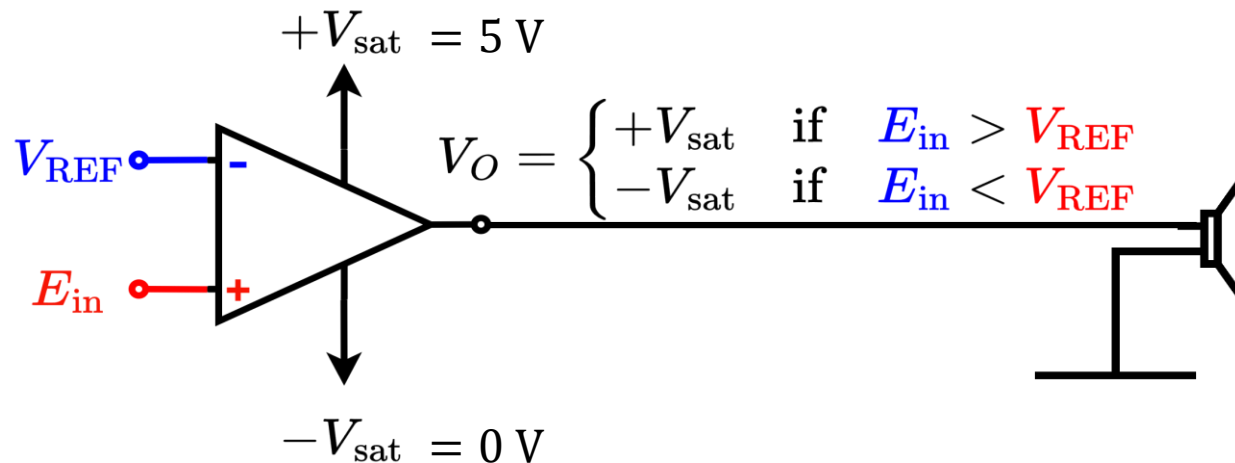
Example 05: Smoke Detector

No Smoke

- No scattering
- Low intensity at LDR (Dark - $I \downarrow$)
- High resistance across LDR ($R_{\text{LDR}} \uparrow$)
- Decrease in E_{in}
- Possibility of ($E_{\text{in}} < V_{\text{REF}}$)
- $V_O = -V_{\text{sat}} \rightarrow$ No Alarm

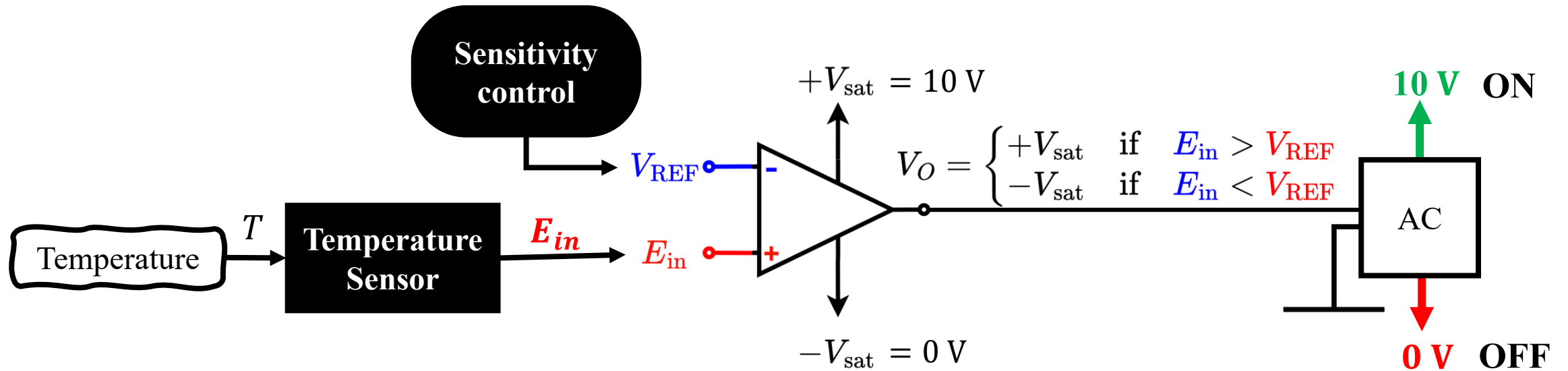
Smoke Present

- High scattering from smoke particles
- High intensity at LDR (Light - $I \uparrow$)
- Low resistance across LDR ($R_{\text{LDR}} \downarrow$)
- Increase in E_{in}
- Possibility of ($E_{\text{in}} > V_{\text{REF}}$)
- $V_O = +V_{\text{sat}} \rightarrow$ Alarm Rings!



**Non-inverting
Op-Amp comparator**

Example 06: Automatic AC



- Let's revisit the example of **AC** to switch it ON based on room **Temperature**.
 When Temp. $\uparrow$; **AC** – ON; Requiring **10V** to turn ON the **AC**
 When Temp. $\downarrow$; **AC** – OFF; Requiring **0V** to turn OFF the **AC**
 Temp $\uparrow \propto E_{in} \uparrow \rightarrow \text{AC} - \text{ON}$ → **NON-INVERTING RELATIONSHIP: NON-INVERTING COMPARATOR**
- The sensitivity of switching on the **AC** is determined by V_{REF}

Comparator Application to Any Device

- Comparators can be used to switch on **any device** based on a **Physical Quantity**.

Phys. Qnty. $\uparrow$ **DEVICE** – ON $\rightarrow$ **Non-Inverting Comparator**

Phys. Qnty. $\downarrow$ **DEVICE** – ON $\rightarrow$ **Inverting Comparator**

Phys. Qnty. $\rightarrow$ $E_{in} \downarrow \uparrow \rightarrow$ **DEVICE** – ON/OFF

This Relationship:
Determines the type of **Comparator**

Determine V_{REF} from the switching condition from the provided information

Physical Quantity

x

Sensor

E_{in}

V_{REF}

V_H

V_L

v_O

Device

ON

OFF

Determine type of comparator from the $E_{in} \rightarrow v_O$ relationship.

Set V_H = Required Voltage to turn ON the device

Set V_L = Minimum Voltage to turn OFF the device